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PVV Score: price / volume / volatility selection for 21-63 day forward risk-adjusted returns

Universe: 503 Finviz S&P 500 constituents (file `data/universe_finviz.csv`), sector mapped to SPDR ETF.

Data: yfinance daily OHLCV, dividend/split adjusted, 2012-2026-10-05. Backtest window 2014-01-01 onward; the 2022-10 onward window ('recent') is the primary evidence per the mandate, the full window is the secondary check.

All statistics are out-of-sample where labelled OOS. Nothing here uses fundamentals, earnings or valuation.

1. Bottom line

- **The ranking is identity-neutral.** Every factor is expressed as the name's deviation from its own trailing 3-year norm (z), so a chronically strong, drawdown-free, high-Sharpe name (DELL, NVDA, LLY, PGR, BRK-B) scores near the middle unless its current price / volume / volatility behaviour is unusual for it. Weights are fitted walk-forward inside beta terciles. Every one of the 503 tickers has a score (section 6); names without 252 sessions of history are listed with the reason.
- **Out of sample (walk-forward 2017-2026), the state composite is the more robust of the two systems.** Recent window: 42d IC 0.039, 63d IC 0.056 (NW t 2.3); beta-neutral 63d IC 0.029 (t 2.4). Full window beta-neutral: 42d IC 0.017 (t 2.6), 63d 0.023 (t 2.9). The level-based composite had a slightly higher recent raw IC (0.047) but almost none of it survived beta-neutralisation (0.006) and half of it was ticker identity (section 7). Top-vs-bottom decile 42d Sharpe spread of the state composite: +0.22 annualised.
- **What precedes top-decile forward risk-adjusted windows, across all names (section 7b):** small but consistent pre-states. Pooled: 63d relative strength unusually WEAK for the name (bottom-quintile lift 1.08, top-quintile 0.85), dollar volume unusually low, no recent ignition day, downside-vol asymmetry unusually high, longer-than-usual time since the 21d high. That is a quiet, pulled-back, under-owned state, i.e. 1-3 month mean reversion, not breakout chasing. Mid-beta names add: 63d base unusually tight (lift 1.06 vs 0.84), downside vol unusually low, a recent ignition day. The identity layer (section 7) rewards persistent strength; the state layer rewards an unusual quiet pullback in a name. Both are real and they are different questions: which names, and when.
- The identity-heavy level composite and the LightGBM case-study model are kept in the files for comparison and are not used in the ranking.
- Compression / NR7 / Bollinger squeeze, ignition-day volume multiples, accumulation-day counts, gap behaviour, streaks and close-location factors showed no pooled 1-3 month signal as LEVELS. As STATES a few re-appear conditionally (tight base and ignition in mid beta); the rest stay flat.

2. Method

- **Targets.** For every (date, stock): realised Sharpe and Sortino of the stock over $t+1..t+h$ ($h = 21, 42, 63$) minus the same for SPY and for the stock's sector ETF (XLRE/XLC spliced to XLF/XLK before inception). Also the information-ratio form (Sharpe of stock-minus-SPY). The ML label is the per-date percentile of SPY-excess Sharpe averaged over the three horizons.
- **Eligibility (point-in-time).** Price $\geq \$5$, 63d median dollar volume $\geq \$10M$, ≥ 252 sessions of history. Survivorship caveat: the universe is today's constituents, so absolute levels are biased up; cross-sectional ranks are far less affected.
- **Factors.** 86 point-in-time factors in five families (price structure, volume, volatility, relative strength, run character) plus two user-described interaction setups. Each verified against hand calculations and a perturbation test for look-ahead (none).
- **Statistics.** Daily cross-sectional Spearman IC; Newey-West t-stats with lag = horizon because forward windows overlap; Benjamini-Hochberg across the 86-factor zoo; decile spreads in raw excess-Sharpe units; 21-day rank autocorrelation (turnover).
- **Conditioning.** Everything re-run inside trailing-beta terciles, 20d realised-vol terciles, sector groups (defensive / cyclical / growth), per sector, and with sector-neutral ranks.
- **Case-study model.** LightGBM trained walk-forward by calendar year (2017-2026), 63-day embargo, 2-year half-life time-decay weights, every 5th day sampled for training. Four pre-specified regularised variants were also run. Nearest-neighbour analogs (50 historical cases, max 3 per ticker) give each live name its realised-outcome evidence.
- **Composite.** Factor selection and weights re-estimated each year on training data only ($|\text{NW } t| \geq 1.5$ on the trailing 3y, sign agreement with full history, greedy de-duplication at $|\rho| > 0.7$, weights = recency-blended IC). Pooled and beta-bucketed variants.

3. Single-factor results (SPY-excess Sharpe, mean IC across 21/42/63d)

Power note: the recent window is $\sim 1,000$ sessions = roughly 24 independent 42-day windows, so $|t|$ around 2 is the attainable ceiling for a genuine IC of 0.05-0.07. **No factor survives Benjamini-Hochberg at 5% in either window.** The evidence is in coherent clusters, not individual rows.

	group	prior	ic_spy_full	t_spy_full	ic_spy_recent	t_spy_recent	ic_sec_recent	t_sec_recent	sign_stable	ic42_pos_years	d10_d1_spread_recent_42	autocorr_21
rs_lead_126	relative	1	-0.002	-0.127	0.066	2.436	0.013	0.705	False	0.538	0.515	0.72
ceiling_2x_low	run	-1	0.01	1.151	0.038	2.275	0.019	1.662	True	0.615	0.28	0.743
corr_spy_63	relative	-1	0.02	1.177	0.072	2.121	-0.001	-0.1	True	0.615	0.437	0.848
parkinson_cc_20	vol	0	-0.004	-0.483	-0.029	-1.977	0.008	1.081	True	0.462	-0.182	0.081
dist_52w_low	run	0	0.002	0.154	0.047	1.915	0.027	1.569	True	0.538	0.417	0.869
up_capture_63	relative	1	0.005	0.219	0.069	1.911	0.001	0.024	True	0.538	0.55	0.817
overnight_intraday_vol_20	vol	0	0.011	1.099	0.034	1.908	-0.001	-0.083	True	0.692	0.211	0.203
beta_63	relative	0	-0.004	-0.23	0.072	1.815	-0.007	-0.404	False	0.538	0.539	0.88
days_since_2x_low	run	0	-0	-0.013	-0.037	-1.791	-0.028	-2.47	True	0.385	-0.262	0.923
gap_share_21	price	0	0.014	1.335	0.037	1.765	0.007	0.623	True	0.692	0.32	0.26

	group	prior	ic_spy_full	t_spy_full	ic_spy_recent	t_spy_recent	ic_sec_recent	t_sec_recent	sign_stable	ic42_pos_years	d10_d1_spread_recent_42	autocorr_21
beta_252	relative	0	0.002	0.06	0.072	1.727	-0.013	-0.66	True	0.462	0.398	0.985
mom_12_1	price	1	0.005	0.374	0.04	1.66	0.043	2.288	True	0.615	0.277	0.892
up_gap_freq_21	price	1	-0.009	-0.532	0.041	1.583	-0.003	-0.188	False	0.538	0.318	0.452
down_capture_63	relative	-1	0.003	0.112	0.051	1.461	-0.02	-1.236	True	0.538	0.308	0.792
dn_vol_63	vol	-1	-0.011	-0.681	0.043	1.457	-0.014	-0.846	False	0.615	0.239	0.845
rv20	vol	0	-0.015	-0.961	0.04	1.37	-0.01	-0.67	False	0.538	0.274	0.545
rs_newhigh_price_not_20	relative	1	-0.006	-0.694	-0.017	-1.362	0	0.046	True	0.692	0.011	0.196
dvol_trend_21_126	volume	0	-0.002	-0.333	-0.015	-1.343	0.011	1.395	True	0.462	-0.087	0.242
vol_confirms_price_21	roc	1	-0.004	-0.896	-0.011	-1.272	-0.016	-2.361	True	0.538	-0.013	0.008
accum_dist_50	volume	1	-0.009	-1.034	-0.019	-1.255	0.006	0.506	True	0.769	-0.173	0.563
rv_ratio_5_20	vol	0	0.001	0.384	0.006	1.188	0.002	0.671	True	0.538	0.026	-0.006
base_tightness_63	price	-1	-0.011	-0.734	0.033	1.136	-0.004	-0.283	False	0.615	0.21	0.752
bbw_squeeze_126	price	-1	-0.004	-0.64	0.015	1.134	0.011	1.316	False	0.538	0.121	0.108
vol_turn_off_dryup	roc	1	0	0.135	0.002	1.118	0.001	0.703	True	0.538	0.033	0.001
rv10_roc_5	roc	0	-0	0.047	0.005	1.093	0.002	0.516	False	0.385	0.011	-0.003

d10_d1_spread_recent_42 is the annualised 42d Sharpe difference between the factor's top and bottom decile (recent window). autocorr_21 near 1 = slow factor; near 0 = daily-turnover factor.

Reading of the clusters

- **Trend-persistence / leadership cluster** (rs_lead_126 , mom_12_1 , dist_52w_low , up_capture_63 , days_since_2x_low negative, ceiling_2x_low positive): the only group with a consistent positive sign in the recent window and a positive sector-excess reading. Note ceiling_2x_low comes in with the OPPOSITE sign to the prior: names already 100% off their 52w low have continued to produce better forward Sharpe in 2022-2026. In the pooled sample the "ceiling" is not a ceiling.
- **Beta / correlation cluster** (beta_63 , beta_252 , corr_spy_63 , rv20 , dn_vol_63): positive in the recent window, flat over the full window, negative vs sector. This is the bull-regime beta bet, not stock selection. The composite and the model are therefore also evaluated beta-neutral (section 5).
- **Compression / dry-up setups** (range_comp_*, bbw_*, nr7_count_20 , vol_dry_*, dryup_x_tight): no pooled signal at these horizons. They show up only conditionally (section 4), in low-vol / defensive names.
- **Ignition / volume-burst factors** (ignition_mult_10 , vol_z_1 , big_vol_updays_net_21 , ignite_x_rs): no signal for 21-63d forward Sharpe in S&P 500 names. These are shorter-horizon or small-cap phenomena.
- **Short-term price factors** (ret_5 , clv_5 , streak_now , pos_in_20d_range): nothing. Daily-turnover factors with no 1-3 month payoff.

3b. Rate-of-change family (added in the rebuild): did it carry signal?

35 ROC factors were added: point-to-point ROC of volume (5/10/21), dollar volume, realised vol (5/10/21, fast 10d RV, Parkinson, range, Bollinger width), downside vol, vol asymmetry, beta, correlation, idiosyncratic vol and vol-of-vol; acceleration (ROC of ROC) for price, volume and RV; and interactions:

volume turning up off a dry-up, RV expanding off compression, volume confirming price, volume rising on a flat tape, volume up while vol down. They went through every gate below alongside the original 86.

Single-factor IC, all 35 ROC factors (SPY-excess Sharpe, mean over 21/42/63d):

	prior	ic_spy_full	t_spy_full	ic_spy_recent	t_spy_recent	ic_sec_recent	t_sec_recent	sign_stable	d10_d1_spread_recent_42	autocorr_21
vol_confirms_price_21	1	-0.004	-0.896	-0.011	-1.272	-0.016	-2.361	True	-0.013	0.008
vol_turn_off_dryup	1	0	0.135	0.002	1.118	0.001	0.703	True	0.033	0.001
rv10_roc_5	0	-0	0.047	0.005	1.093	0.002	0.516	False	0.011	-0.003
vol_cv_roc_10	0	-0.003	-1.036	-0.006	-1.08	-0.005	-1.245	True	-0.038	-0.277
dnvol_roc_21	-1	-0.001	-0.063	0.012	0.984	0.009	1.001	False	0.049	0.047
vov_roc_21	0	-0.002	-0.384	-0.009	-0.984	-0.002	-0.151	True	-0.048	0.009
rv20_roc_10	0	-0	-0.031	0.006	0.8	0.004	0.845	False	-0.001	-0.377
vol_up_rv_down	1	-0.001	-0.208	-0.004	-0.75	-0.004	-0.836	True	0.012	-0.092
px_accel_21	0	-0.003	-0.424	0.01	0.734	-0.006	-0.696	False	0.164	-0.472
rv20_roc_5	0	-0.001	-0.225	0.004	0.727	0.004	0.925	False	0.021	-0.323
rs_slope_21	0	-0.003	-0.425	0.01	0.726	-0.006	-0.685	False	0.164	-0.472
corr_roc_21	0	-0.013	-1.559	0.009	0.702	0.006	0.623	False	0.091	0.043
vol_roc_x_rv_roc	0	-0.003	-0.963	0.003	0.693	-0.002	-0.589	False	-0.027	0.018
nvf_roc_21	1	-0.006	-0.762	-0.011	-0.689	-0.006	-0.549	True	-0.084	-0.016
idio_roc_21	0	-0.003	-0.465	-0.008	-0.687	-0.003	-0.36	True	-0.04	0.017
vol_roc_5	0	-0.001	-0.25	0.002	0.6	-0	-0.109	False	-0.005	0.008
updown_vol_roc_21	1	-0.002	-0.337	0.006	0.576	-0.004	-0.521	False	0.143	-0.482
bbw_roc_10	0	-0.002	-0.535	0.004	0.565	-0.001	-0.137	False	0.055	-0.152
vol_dry_roc_5	0	-0.002	-0.541	0.002	0.555	-0.002	-0.461	False	0.006	-0.037
dist_high_roc_10	1	-0.008	-1.265	-0.006	-0.514	-0.009	-1.275	True	0.003	-0.038
rv_turn_off_compression	1	-0.001	-0.276	0.001	0.48	0.002	0.726	False	0.041	-0.066
dvol_roc_21	0	-0.001	-0.16	-0.004	-0.458	0.002	0.335	True	-0.016	-0.344
px_accel_5	0	-0.001	-0.41	-0.002	-0.441	-0.004	-1.452	True	-0.034	-0.003
rv_accel_5	0	-0.001	-0.444	0.001	0.399	-0.001	-0.274	False	-0.003	-0.282
beta_roc_21	0	-0.018	-1.863	-0.005	-0.352	0.002	0.206	True	-0.018	0.078
rv20_roc_21	0	0	0.063	0.002	0.303	0.002	0.245	True	-0.005	-0.456
range_roc_10	0	-0.002	-0.653	0.001	0.262	0.003	0.606	False	-0.005	-0.014
vol_roc_21	0	0.002	0.426	-0.002	-0.241	0.001	0.104	False	-0.008	-0.345
accum_roc_21	1	-0.009	-1.33	-0.004	-0.227	-0.002	-0.194	True	-0.038	-0.001
asym_roc_21	-1	0.004	0.457	0.003	0.168	0.01	1.033	True	0.014	-0.012
range_pos_roc_10	0	-0.004	-0.813	-0.001	-0.124	-0.006	-1.041	True	0.033	-0.174
vol_accel_5	0	-0.001	-0.495	0	0.116	-0.002	-1.26	False	0.016	0.019
vol_roc_10	0	-0.001	-0.456	-0	-0.081	-0.002	-0.354	True	-0.014	-0.051
quiet_accum_21	1	0.003	0.659	-0.001	-0.076	-0	-0.005	False	-0.026	-0.29
park_roc_10	0	-0.001	-0.287	0	0.064	0.001	0.292	False	-0.012	-0.394

Conditional (recent, 42d, NW t) for the ROC factors, top 15 by max |t|:

	beta_low	beta_mid	beta_high	vol_low	vol_mid	vol_high	grp_defensive	grp_cyclical	grp_growth
vol_confirms_price_21	-0.47	-1.28	-0.84	-1.12	-3.5	0.19	-0.2	-2.22	-0.56
vov_roc_21	0.14	0.5	-1.3	-0.64	0.41	-3.3	0.89	-2.33	-0.8
vol_roc_x_rv_roc	2.71	-0.66	-1.27	1.87	1.11	-1.51	2.24	0.25	-2.1

	beta_low	beta_mid	beta_high	vol_low	vol_mid	vol_high	grp_defensive	grp_cyclical	grp_growth
rv20_roc_5	2.55	-0.12	0.01	0.86	0.3	-1.02	2.05	0.51	-0.37
rv20_roc_10	2.41	-0.2	0.05	0.42	0.17	-1.48	1.88	0.48	-0.19
idio_roc_21	0.23	-0.77	-0.94	-0.33	-0.21	-1.86	2.41	-1.07	-1.7
rv_turn_off_compression	1.3	0.22	0.84	-0.18	0.38	-0.11	2.39	1.56	-0.8
rv20_roc_21	1.76	-0.42	-1.07	0.25	-0.21	-2.34	1.08	0.18	-0.72
vol_up_rv_down	-0.89	-1.01	0.25	-0.97	-0.55	2.34	-1.15	-0.37	0.76
rv10_roc_5	2.32	-0.75	0.73	1.36	1	1.42	1.11	0.45	1.13
vol_cv_roc_10	-1	-2.04	0.09	-0.64	-2.23	-0.66	-0.13	-1.85	-0.45
range_roc_10	1.74	-0.81	-2	0.17	0.01	-0.42	1	-0.58	-0.27
vol_roc_21	1.72	-1.48	0.57	0.24	-0.47	-0.82	1.99	-0.19	0.24
dnvol_roc_21	1.03	-0.75	0.15	0.22	-0.2	-1.8	1.59	-0.56	-1.04
park_roc_10	1.78	-0.38	-1.72	-0.03	-0.33	-1.5	1.46	-0.39	-0.8

Event study, state version of the ROC factors, 5 sessions before top-decile windows (recent, pooled), top 15:

factor	mean_rank_excess	t_cluster	n_clusters	lift_top_q	lift_bottom_q
z_vol_turn_off_dryup	-0.011	-2.021	2802	0.943	1.044
z_vol_confirms_price_21	-0.01	-1.893	2807	0.963	1.014
z_corr_roc_21	0.009	1.671	2804	0.917	1.045
z_range_pos_roc_10	0.008	1.574	2808	0.991	0.981
z_vol_dry_roc_5	-0.008	-1.509	2805	0.985	0.978
z_vol_roc_5	-0.008	-1.466	2809	0.965	0.964
z_idio_roc_21	-0.008	-1.414	2804	0.978	0.991
z_px_accel_21	0.007	1.295	2807	0.987	0.939
z_rv_accel_5	-0.007	-1.254	2808	0.953	0.953
z_vov_roc_21	-0.007	-1.213	2803	0.944	0.932
z_vol_up_rv_down	-0.007	-1.194	2808	0.986	0.993
z_accum_roc_21	-0.006	-1.153	2804	0.999	1.009
z_rs_slope_21	0.006	1.145	2807	0.98	0.939
z_vol_cv_roc_10	0.005	0.98	2808	0.98	1.024
z_dvol_roc_21	-0.005	-0.843	2807	0.937	1.049

Walk-forward OOS composites, before vs after adding the ROC family (beta-bucketed variants; same folds, same rules):

composite	IC 42d full	t 42d full	IC 42d recent	t 42d recent	IC 42d beta-neutral recent	IC 63d full	t 63d full	IC 63d recent	t 63d recent	IC 63d beta-neutral recent
state, before ROC	0.019	1.453	0.039	1.597	0.015	0.022	1.552	0.056	2.28	0.029
state, with ROC	0.02	1.771	0.025	1.278	0.013	0.024	1.813	0.041	1.961	0.026
level, before ROC	0.018	1.086	0.047	1.463	0.006	0.025	1.346	0.065	1.905	0.015
level, with ROC	0.017	1.076	0.044	1.485	0.008	0.024	1.392	0.064	2.058	0.018

Share of absolute weight the state composite gave to ROC factors, by fold and bucket:

year	high	low	mid	pooled
2017	0.35	0.09	0.13	0.14
2018	0.28	0.3	0.17	0.22
2019	0.36	0.31	0.15	0.29

	year	high	low	mid	pooled
	2020	0.31	0.33	0.23	0.41
	2021	0.29	0.02	0.28	0.22
	2022	0.24	0.05	0.16	0.09
	2023	0.06	0.11	0.08	0.02
	2024	0.24	0.15	0.11	0.16
	2025	0.44	0.18	0.24	0.11
	2026	0.15	0.29	0.31	0.14

ROC factors the state composite selected, and in how many of the 10 folds:

factor	folds	mean_w
z_vol_turn_off_dryup	8	-0.065
z_vol_roc_x_rv_roc	8	-0.026
z_rv20_roc_5	7	0.021
z_rv20_roc_10	7	0.024
z_rv_turn_off_compression	6	0.005
z_rv20_roc_21	6	0.009
z_corr_roc_21	5	-0.083
z_park_roc_10	5	0.009
z_rs_slope_21	5	-0.061
z_rv10_roc_5	4	0.022
z_updown_vol_roc_21	4	-0.045
z_idio_roc_21	4	-0.042
z_range_roc_10	4	-0.005
z_range_pos_roc_10	4	-0.046
z_vol_cv_roc_10	4	-0.024
z_accum_roc_21	3	-0.062
z_asym_roc_21	3	0.051
z_vol_confirms_price_21	3	-0.037
z_vol_roc_21	3	0.03
z_quiet_accum_21	3	0.003

(Reading of these tables is in section 8.)

4. Conditional results: where each factor works (recent window, 42d, NW t-stats)

	beta_low	beta_mid	beta_high	vol_low	vol_mid	vol_high	grp_defensive	grp_cyclical	grp_growth	sector_neutral_vs_sector
mom_12_1	1.41	1.14	3.59	2.06	2.54	2.02	0.99	1.72	0.82	1.79
obv_price_div_63	0.5	-0.34	0.92	0.36	0.91	-0.72	3.56	-0.67	-0.8	0.32
vol_confirms_price_21	-0.47	-1.28	-0.84	-1.12	-3.5	0.19	-0.2	-2.22	-0.56	nan
vov_roc_21	0.14	0.5	-1.3	-0.64	0.41	-3.3	0.89	-2.33	-0.8	nan
dvol_trend_21_126	-2.93	-0.26	1.45	-1.76	-1.23	-0.49	-1.3	-1.36	1.16	0.2
skew_63	0.44	0.36	-2.09	-0.16	-0.92	1.02	-2.24	0.56	2.88	0.45
corr_spy_63	0.83	2.74	1.18	2.09	1.83	1.85	0.72	1.83	0.43	1.72
vol_roc_x_rv_roc	2.71	-0.66	-1.27	1.87	1.11	-1.51	2.24	0.25	-2.1	nan
ceiling_2x_low	1.61	-1.08	2.7	0.76	2.17	2.22	-0.02	1.02	2.58	2.2

	beta_low	beta_mid	beta_high	vol_low	vol_mid	vol_high	grp_defensive	grp_cyclical	grp_growth	sector_neutral_vs_sector
idio_vol_63	-1.1	-2.4	0.47	-2.12	-2.69	1.03	-0.04	-1.57	1.75	-0.06
days_since_2x_low	-0.1	0.79	-2.67	-1.08	-1.62	-1.51	0.13	-0.82	-2.25	-2.54
rs_lead_126	1.56	1.97	1.76	2.59	1.68	2.35	1.31	2.43	1.18	2.11
rv20_roc_5	2.55	-0.12	0.01	0.86	0.3	-1.02	2.05	0.51	-0.37	nan
capture_spread_63	-0.93	1.67	1.7	-0.41	0.22	1.39	-1.64	-0.07	2.54	0.36
days_since_52w_high	-1.11	-0.62	-2.52	-0.53	-1.43	-1.55	-0.42	-0.63	-0.6	-0.75
dist_52w_low	0.85	1.59	2.5	1.9	2.11	2.01	0.31	1.48	1.82	1.79
up_day_frac_63	-0.94	0.86	2.43	0.23	0.14	0.27	-0.11	-1.09	0.76	-0.1
dn_up_vol_asym_63	0.73	-1.16	0.5	0.5	0.98	-0.91	2.33	1.07	-2.42	0.57
rv20_roc_10	2.41	-0.2	0.05	0.42	0.17	-1.48	1.88	0.48	-0.19	nan
idio_roc_21	0.23	-0.77	-0.94	-0.33	-0.21	-1.86	2.41	-1.07	-1.7	nan
rv_turn_off_compression	1.3	0.22	0.84	-0.18	0.38	-0.11	2.39	1.56	-0.8	nan
rs_sec_63	-0.33	2.39	1.33	0.75	0.24	1.28	-0.84	-0.76	1.47	-0.4
base_depth_126	-0.14	2.39	1.1	0.42	0.85	0.36	-0.86	0.34	0.85	0.28
rs_sec_21	1.19	2.36	1.73	1.77	1.85	2	0.43	0.71	1.71	0.91
rv20_roc_21	1.76	-0.42	-1.07	0.25	-0.21	-2.34	1.08	0.18	-0.72	nan
vol_up_rv_down	-0.89	-1.01	0.25	-0.97	-0.55	2.34	-1.15	-0.37	0.76	nan
rs_dist_high_126	-0.17	2.32	1.18	1.57	1.1	0.65	-0.88	1.16	0.98	0.75
rv10_roc_5	2.32	-0.75	0.73	1.36	1	1.42	1.11	0.45	1.13	nan
rv_ratio_20_60	1.48	-0.72	-0.87	0.33	-0.16	-2.3	0.94	0.63	-1.08	-0.07
vol_cv_roc_10	-1	-2.04	0.09	-0.64	-2.23	-0.66	-0.13	-1.85	-0.45	nan

Multiple-testing note: 86 factors x 10 buckets = 860 tests; the expected maximum $|t|$ under the null is about 3.2, so single cells near 3 are not individually reliable. What is reliable is the pattern:

- **High beta / growth:** the trend-persistence cluster is strongest here (`mom_12_1` $t=3.6$ in the high-beta tercile, `days_since_52w_high` -2.5, `ceiling_2x_low` +2.7, `days_since_2x_low` -2.7). In high-beta names, buy strength, recent 2x-off-low crossings, and proximity to highs.
- **Low beta / low vol / defensive:** the sign of several price factors flips or dies, and volume-structure factors appear instead: `obv_price_div_63` +3.6 in defensives (volume leading price), `vol_dry_20_250` -2.1 in the low-vol tercile (dry-up works for utilities/staples-type names), `dvol_trend_21_126` -2.9 in low beta (quiet names outperform), `idio_vol_63` negative in low/mid beta. This is the DUK-vs-NVDA asymmetry the mandate anticipated, and it is why the composite is fitted per beta bucket.
- **Per sector** (`conditional_ic_42_per_sector.csv`): indicative only (18-80 names per date). Notable: Utilities `rs_lead_126` +3.1 and dry-up negative; Consumer Defensive `ceiling_2x_low` -3.2 (the ceiling DOES hold in staples) and compression factors positive; Financials show reversal (`ret_63` -2.3).

5. Out-of-sample predictive models (walk-forward 2017-2026)

5a. Case-study model (LightGBM) vs single-factor baselines

lightgbm_walkforward

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.01	1.094	0.012	1.624	0.01	1.38
(21, 'recent')	-0	-0.015	0.004	0.349	0.017	1.72
(42, 'full')	0.013	1.066	0.014	1.56	0.014	1.624
(42, 'recent')	0.003	0.198	0.003	0.204	0.013	1.071
(63, 'full')	0.018	1.21	0.017	1.844	0.016	1.641
(63, 'recent')	0.018	0.884	0.009	0.788	0.009	0.613

baseline_mom_12_1

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.018	1.347	0.026	2.286	0.013	1.325
(21, 'recent')	0.038	1.918	0.044	2.534	0.037	2.337
(42, 'full')	0.015	0.847	0.027	1.879	0.008	0.639
(42, 'recent')	0.041	1.598	0.048	2.361	0.043	2.097
(63, 'full')	0.014	0.659	0.03	1.93	0.006	0.414
(63, 'recent')	0.041	1.464	0.054	2.609	0.051	2.429

baseline_rs_lead_126

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.009	0.718	0.006	0.64	-0.001	-0.085
(21, 'recent')	0.048	2.261	0.026	2.058	0.005	0.448
(42, 'full')	0.015	0.897	0.006	0.524	-0	-0.007
(42, 'recent')	0.068	2.393	0.034	2.057	0.015	0.853
(63, 'full')	0.017	0.92	0.007	0.548	-0.001	-0.079
(63, 'recent')	0.082	2.654	0.041	2.237	0.017	0.814

baseline_beta_252

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.008	0.415	0.001	0.508	-0.019	-2.127
(21, 'recent')	0.05	1.538	-0.004	-1.427	-0.013	-0.837
(42, 'full')	0.02	0.775	0.004	1.248	-0.015	-1.164
(42, 'recent')	0.076	1.734	-0.003	-0.859	-0.011	-0.537
(63, 'full')	0.027	0.868	0.003	0.978	-0.014	-0.896
(63, 'recent')	0.089	1.909	-0.006	-1.307	-0.014	-0.608

LightGBM 42d IC by year: 2017: -0.011, 2018: +0.037, 2019: +0.141, 2020: +0.006, 2021: -0.005, 2022: -0.042, 2023: -0.037, 2024: +0.006, 2025: +0.047, 2026: -0.021

Regularised variants (rank-only features, 7 leaves, 300 rounds; ridge; with/without decay; no market-context features):

gbm_reg_rankonly_decay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.004	0.384	0.002	0.242	0.008	1.332
(21, 'recent')	0.002	0.116	-0.004	-0.385	0.001	0.185
(42, 'full')	0.01	0.687	0.006	0.57	0.013	1.496

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(42, 'recent')	0.011	0.507	-0.005	-0.341	0.002	0.198
(63, 'full')	0.013	0.803	0.009	0.923	0.018	1.688
(63, 'recent')	0.02	0.793	0.001	0.053	0.002	0.239

gbm_reg_rankonly_noddecay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.005	0.507	0.005	0.661	0.01	1.504
(21, 'recent')	-0.01	-0.714	-0.009	-0.863	-0.002	-0.275
(42, 'full')	0.012	0.867	0.011	1.126	0.016	1.809
(42, 'recent')	-0	-0.018	-0.006	-0.516	-0.001	-0.077
(63, 'full')	0.017	1.1	0.016	1.501	0.023	1.997
(63, 'recent')	0.013	0.677	0.002	0.153	0.002	0.22

ridge_rank_decay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.009	0.813	0.006	0.73	0.014	2.108
(21, 'recent')	0.01	0.519	0	0.024	0.01	1.272
(42, 'full')	0.013	0.868	0.006	0.543	0.013	1.409
(42, 'recent')	0.021	0.826	-0.002	-0.147	0.003	0.242
(63, 'full')	0.016	0.947	0.009	0.778	0.015	1.262
(63, 'recent')	0.029	1.032	0.001	0.07	-0.007	-0.531

gbm_reg_noctx_decay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.007	0.633	0.008	0.957	0.005	0.818
(21, 'recent')	0.002	0.139	0.001	0.112	0.012	1.501
(42, 'full')	0.01	0.718	0.009	0.868	0.01	1.184
(42, 'recent')	0.012	0.532	0.001	0.045	0.015	1.389
(63, 'full')	0.014	0.903	0.011	1.137	0.015	1.425
(63, 'recent')	0.025	1.027	0.006	0.469	0.017	1.374

5b. Walk-forward linear composite (LEVEL factors; identity-heavy, superseded by section 6)

composite_pooled_wf_level

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.002	0.168	-0.001	-0.153	0.007	1.014
(21, 'recent')	-0.014	-0.802	-0.024	-2.019	-0.011	-1.13
(42, 'full')	0.005	0.314	-0.001	-0.086	0.006	0.6
(42, 'recent')	-0.007	-0.312	-0.032	-2.575	-0.017	-1.362
(63, 'full')	0.012	0.651	0.004	0.329	0.008	0.641
(63, 'recent')	0.015	0.633	-0.021	-1.44	-0.017	-1.428

composite_betabucket_wf_level

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.008	0.648	0.005	0.674	-0.002	-0.268
(21, 'recent')	0.024	1.078	0.004	0.29	0.008	0.572
(42, 'full')	0.017	1.076	0.007	0.735	0	0.029
(42, 'recent')	0.044	1.485	0.008	0.551	0.012	0.679
(63, 'full')	0.024	1.392	0.009	0.844	0.002	0.183
(63, 'recent')	0.064	2.058	0.018	1.139	0.015	0.729

6. Live scoring: full universe, identity-neutral

6a. Walk-forward OOS: state composite vs level composite

composite_pooled_wf_state

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.001	0.133	0.001	0.082	-0.004	-0.603
(21, 'recent')	-0.002	-0.123	-0.01	-0.872	-0.004	-0.338
(42, 'full')	0.012	0.988	0.006	0.623	-0.007	-0.792
(42, 'recent')	0.006	0.315	-0.012	-0.851	-0.011	-0.654
(63, 'full')	0.016	1.247	0.007	0.713	-0.007	-0.769
(63, 'recent')	0.012	0.537	-0.008	-0.464	-0.017	-1.027

composite_betabucket_wf_state

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.01	1.16	0.006	1.166	0.001	0.155
(21, 'recent')	0.013	0.883	0.005	0.554	0.014	1.537
(42, 'full')	0.02	1.771	0.016	2.368	0.005	0.632
(42, 'recent')	0.025	1.278	0.013	1.241	0.018	1.486
(63, 'full')	0.024	1.813	0.02	2.601	0.006	0.698
(63, 'recent')	0.041	1.961	0.026	2.166	0.023	1.725

Identity decomposition of the state composite (recent window, 42d): raw IC 0.039, persistent 0.027, deviation 0.039; correlation of today's rank with the name's own historical-average rank 0.42 (level composite: 0.58). The signal now sits in the deviation component.

6b. Final state-composite weights by beta bucket (z = deviation from the name's own 3y norm)

	low	mid	high
z_amihud_21	nan	nan	-0.15
z_base_depth_126	nan	0.144	nan
z_clv_21	nan	0.107	nan
z_corr_spy_63	nan	0.132	nan
z_days_since_20pct_dd	0.264	0.126	nan
z_days_since_2x_low	-0.285	nan	-0.065
z_days_since_52w_high	nan	nan	-0.098

	low	mid	high
z_days_since_ignition	nan	nan	0.072
z_dn_vol_63	nan	nan	0.084
z_dvol_trend_21_126	-0.209	nan	nan
z_ignition_mult_10	nan	nan	-0.053
z_jump_21	nan	nan	-0.048
z_log_dvol_63	nan	0.133	0.153
z_mom_12_1	nan	nan	0.157
z_nr7_count_20	-0.112	nan	nan
z_parkinson_cc_20	nan	nan	0.046
z_rs_sec_63	nan	0.143	nan
z_sharpe_21	nan	0.113	nan
z_streak_now	-0.13	nan	nan
z_up_day_frac_63	nan	0.102	0.074

6c. All 503 tickers as of 2026-10-05

state_score_pooled = universe percentile of the state composite (the single sortable number).

state_score = percentile inside the name's beta bucket. level_score = the identity-heavy composite, for comparison. z columns are standard deviations vs the name's own history. top_states = the three readings contributing most to the score today. Sorted by state_score_pooled. The interactive version (filter, sort) is [universe_scores.html](#).

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
ILMN	Healthcare	mid	1	1	0.97	2.83	0.76	1.57	0.94	3.02	-3.04	-1.22	log_dvol_63 z=+3.1; base_depth_126 z=+1.2; rs_sec_63 z=+2.0	nan
EXPD	Industrials	mid	1	0.99	1	2.25	0.35	1.56	-0.18	-0.68	0.9	-1.04	base_depth_126 z=+1.5; rs_sec_63 z=+2.3; corr_spy_63 z=+0.1	nan
WAT	Healthcare	mid	1	0.99	0.98	1.26	0.37	1.33	-0.58	-0.79	1.24	-0.59	base_depth_126 z=+1.2; corr_spy_63 z=+0.1; log_dvol_63 z=+1.6	nan
AWK	Utilities	low	0.99	1	0.92	0.63	-0.06	0.28	-1.36	-0.71	-1.47	-0.33	days_since_20pct_dd z=+1.4; dvol_trend_21_126 z=-0.8; nr7_count_20 z=-1.1	nan
MRNA	Healthcare	high	0.99	1	0.93	5.13	0.51	1.76	-0.28	0.17	-1.38	-2.37	mom_12_1 z=+5.1; log_dvol_63 z=+6.0; amihud_21 z=-4.3	nan
D	Utilities	low	0.99	0.99	0.96	0.18	0.07	-0.78	-0.72	-0.6	0.21	1.58	days_since_20pct_dd z=+1.2; dvol_trend_21_126 z=-1.0; streak_now z=-0.5	nan
WBD	Communication Services	mid	0.99	0.98	0.96	0.03	-0.21	1.42	2.06	3.12	1.67	-1.61	base_depth_126 z=+1.4; days_since_20pct_dd z=+2.7; clv_21 z=+0.6	nan
REG	Real Estate	low	0.99	0.99	0.98	-0.2	-0.29	-2.12	-1.47	-0.05	-0.25	1.38	dvol_trend_21_126 z=-1.0; streak_now z=-2.2; days_since_20pct_dd z=+0.6	nan
ETR	Utilities	low	0.98	0.98	0.8	-0.5	-1.34	-2.04	-0.57	0.4	-0.42	2.08	days_since_20pct_dd z=+1.0; dvol_trend_21_126 z=-0.6; nr7_count_20 z=-1.1	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
MRVL	Technology	high	0.98	0.99	0.89	1.35	-0.44	0.41	-0.93	-1.19	0.27	0.09	log_dvol_63 z=+2.1; amihud_21 z=-2.1; mom_12_1 z=+1.4	nan
SJM	Consumer Defensive	low	0.98	0.98	0.71	2.63	1.29	0.1	-1.07	-1.06	0.8	-0.48	days_since_20pct_dd z=+1.3; nr7_count_20 z=-1.5; dvol_trend_21_126 z=-0.5	nan
IEX	Industrials	mid	0.98	0.98	0.99	2.15	0.34	1.28	-0.94	-0.9	1.11	-0.2	rs_sec_63 z=+2.1; base_depth_126 z=+1.1; sharpe_21 z=+1.2	nan
MU	Technology	high	0.98	0.99	0.78	1.28	-1.38	0.19	-1.45	-1.53	0.39	0.18	amihud_21 z=-2.2; log_dvol_63 z=+1.9; mom_12_1 z=+1.3	nan
PPL	Utilities	low	0.97	0.97	0.95	-1.07	-0.98	-1.92	-1.11	-0.84	-1.38	0.73	dvol_trend_21_126 z=-1.3; days_since_20pct_dd z=+1.0; nr7_count_20 z=-0.9	nan
MTD	Healthcare	mid	0.97	0.97	0.9	0.24	0.66	1.75	0.17	0.68	-0.58	-1.16	base_depth_126 z=+1.4; rs_sec_63 z=+1.8; log_dvol_63 z=+1.9	nan
AEP	Utilities	low	0.97	0.96	0.88	-0.18	-0.75	-1.3	-1.31	0.17	1.21	2.7	days_since_20pct_dd z=+1.1; dvol_trend_21_126 z=-0.6; nr7_count_20 z=-1.0	nan
HPE	Technology	high	0.97	0.98	0.98	2.54	0.2	1.23	0.95	-0.37	-0.31	-0.67	mom_12_1 z=+2.5; log_dvol_63 z=+2.2; amihud_21 z=-1.7	nan
NDSN	Industrials	mid	0.97	0.96	0.99	1.42	0.41	1.26	-0.63	-0.68	-0.7	-1.28	rs_sec_63 z=+3.6; base_depth_126 z=+1.0; corr_spy_63 z=-2.2	nan
FRT	Real Estate	low	0.96	0.96	0.64	0.97	-0.01	-1.27	-0.98	0.87	0.54	1.72	days_since_20pct_dd z=+1.5; streak_now z=-1.0; nr7_count_20 z=-1.0	nan
NTAP	Technology	high	0.96	0.98	1	1	0.16	1	-0.04	-0.07	1.29	-0.17	log_dvol_63 z=+2.6; mom_12_1 z=+1.0; days_since_52w_high z=-1.1	nan
STT	Financial	high	0.96	0.97	0.91	1.98	0.04	-0.32	-0.93	0.58	-0.21	0.37	amihud_21 z=-2.8; mom_12_1 z=+2.0; log_dvol_63 z=+1.8	nan
STLD	Basic Materials	high	0.96	0.96	0.81	1.55	-0.52	-0.47	-0.26	-0.75	0.73	1.1	mom_12_1 z=+1.5; days_since_ignition z=+1.5; log_dvol_63 z=+1.4	nan
WST	Healthcare	mid	0.96	0.96	0.94	0.77	0.4	1.63	-0.57	-0.78	-0.02	0.16	base_depth_126 z=+1.4; sharpe_21 z=+1.5; days_since_20pct_dd z=+2.4	nan
FAST	Industrials	mid	0.95	0.95	0.95	-1.31	-0.75	0.64	-1.52	-0.61	-0.67	0.06	log_dvol_63 z=+1.7; rs_sec_63 z=+1.7; sharpe_21 z=+0.5	nan
VTRS	Healthcare	mid	0.95	0.95	0.87	1.59	-0.22	1.1	-0.99	-0.34	-0.51	0.56	days_since_20pct_dd z=+2.7; log_dvol_63 z=+1.8; base_depth_126 z=+0.9	nan
BBY	Consumer Cyclical	mid	0.95	0.94	0.84	0.8	0.2	0.77	-0.58	-1.24	0.03	0.07	up_day_frac_63 z=+0.9; rs_sec_63 z=+1.2; sharpe_21 z=+0.1	nan
LITE	Technology	high	0.95	0.96	0.92	0.55	-0.19	1.17	-0.97	-0.79	1.1	-0.31	amihud_21 z=-1.9; log_dvol_63 z=+1.6; mom_12_1 z=+0.6	nan
ON	Technology	high	0.95	0.95	0.67	1.05	0.3	-0.53	0	0.56	1.07	0.14	mom_12_1 z=+1.0; log_dvol_63 z=+1.5; days_since_52w_high z=-0.9	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
DOV	Industrials	high	0.94	0.95	0.37	0.09	-0.86	-1.16	-1	0.56	0.38	1.87	amihud_21 z=-1.9; log_dvol_63 z=+1.4; dn_vol_63 z=+1.1	nan
FE	Utilities	low	0.94	0.95	0.9	-0.27	-0.9	-2	-0.51	-0.39	-0.67	0.94	days_since_20pct_dd z=+1.0; dvol_trend_21_126 z=-0.7; streak_now z=+0.3	nan
EXPE	Consumer Cyclical	high	0.94	0.94	0.73	0.41	-0.06	-1.17	-0.5	0.88	0.11	0.48	amihud_21 z=-2.1; log_dvol_63 z=+1.6; days_since_ignition z=+1.4	nan
VZ	Communication Services	low	0.94	0.95	0.87	1.03	0.05	-1.05	-0.4	-0.96	-0.05	-0.38	days_since_20pct_dd z=+1.0; streak_now z=-0.9; dvol_trend_21_126 z=-0.7	nan
TMO	Healthcare	mid	0.94	0.93	0.92	1.06	0.47	1.34	-0.34	0.2	0.39	-1.65	rs_sec_63 z=+2.8; base_depth_126 z=+1.2; corr_spy_63 z=-2.3	nan
ED	Utilities	low	0.93	0.94	0.99	0.91	-0.89	-0.5	-1.12	-0.18	1.02	1.31	streak_now z=-0.9; days_since_20pct_dd z=+0.6; dvol_trend_21_126 z=-0.5	nan
ZBRA	Technology	high	0.93	0.93	0.78	0.67	0.29	1.39	-0.95	-1.35	-1	-2.56	amihud_21 z=-1.8; mom_12_1 z=+0.7; dn_vol_63 z=-1.0	nan
MET	Financial	mid	0.93	0.93	0.98	0.88	0.06	0.65	-0.74	-0.48	0.82	-1.51	rs_sec_63 z=+1.8; log_dvol_63 z=+1.6; days_since_20pct_dd z=+1.9	nan
PFE	Healthcare	mid	0.93	0.92	0.83	1.23	0.15	0.97	-1.5	-1.25	0.31	-1.9	days_since_20pct_dd z=+2.5; rs_sec_63 z=+1.7; base_depth_126 z=+0.7	nan
HPQ	Technology	mid	0.93	0.92	0.7	1.32	0.52	0.61	0.15	-0.8	-0.4	-0.95	log_dvol_63 z=+2.3; rs_sec_63 z=+2.0; sharpe_21 z=-0.0	nan
PTC	Technology	mid	0.92	0.91	0.91	-1.53	0.69	0.67	4.41	4.22	-2.06	-2.44	rs_sec_63 z=+3.5; base_depth_126 z=+1.1; sharpe_21 z=+0.8	nan
DDOG	Technology	high	0.92	0.93	0.94	0.45	0.14	1.06	-1.06	-1.14	1.17	0.84	log_dvol_63 z=+1.8; dn_vol_63 z=+2.2; mom_12_1 z=+0.5	nan
VEEV	Healthcare	mid	0.92	0.9	0.95	-0.56	0.38	0.65	-1.09	-1.28	0.07	-1.18	rs_sec_63 z=+2.3; base_depth_126 z=+0.9; up_day_frac_63 z=+1.8	nan
LNT	Utilities	low	0.92	0.93	0.58	-0.59	-0.18	-2.88	-0.27	0.5	0.04	2.1	days_since_20pct_dd z=+0.8; streak_now z=-0.5; nr7_count_20 z=-0.9	nan
BIIB	Healthcare	mid	0.92	0.9	0.92	1.62	0.81	1.46	-1.45	-1.05	-0.68	-0.26	base_depth_126 z=+1.2; log_dvol_63 z=-1.1; days_since_20pct_dd z=+3.0	nan
EQIX	Real Estate	mid	0.91	0.89	0.86	1.57	-1.08	0.23	0.16	-0.71	0.85	0.02	corr_spy_63 z=+0.1; rs_sec_63 z=+0.9; log_dvol_63 z=+1.3	nan
BF-B	Consumer Defensive	low	0.91	0.93	0.57	1.99	-1.02	1.47	-1.34	-1.65	-2.42	-1.31	days_since_20pct_dd z=+5.1; dvol_trend_21_126 z=-0.9; streak_now z=+1.0	nan
TECH	Healthcare	mid	0.91	0.88	0.93	1.24	-0.27	1.89	-2.31	-0.39	0.66	-0.8	base_depth_126 z=+1.5; log_dvol_63 z=+2.4; sharpe_21 z=+0.3	nan
FTNT	Technology	high	0.91	0.92	0.96	1.66	0.38	1.47	-0.4	-1.24	-0.66	-0.46	mom_12_1 z=+1.7; log_dvol_63 z=+1.8; amihud_21 z=-1.8	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
IVZ	Financial	high	0.91	0.92	0.85	0.65	0.21	0.21	-1.14	-0.7	-0.89	-0.7	amihud_21 z=-1.9; mom_12_1 z=+0.6; log_dvol_63 z=+1.5	nan
CSCO	Technology	high	0.9	0.91	0.89	1.53	-0.01	-0.76	-0.27	-1.24	0.93	0.37	mom_12_1 z=+1.5; log_dvol_63 z=+1.7; dn_vol_63 z=+1.3	nan
DELL	Technology	high	0.9	0.9	0.95	2.27	-0.29	0.81	-0.36	-0.19	0.87	-0.14	mom_12_1 z=+2.3; amihud_21 z=-2.9; log_dvol_63 z=+2.2	nan
CRL	Healthcare	high	0.9	0.9	0.95	2.4	0.59	1.74	-0.42	-0.08	-1.21	-1.99	mom_12_1 z=+2.4; log_dvol_63 z=+2.4; amihud_21 z=-2.3	nan
RVTY	Healthcare	high	0.9	0.89	0.9	2.52	0.99	1.81	0.41	2.04	-1.1	-1.86	mom_12_1 z=+2.5; log_dvol_63 z=+3.0; amihud_21 z=-2.6	nan
T	Communication Services	low	0.9	0.92	0.52	-0.72	-1.46	-0.85	-0.5	-0.98	0.23	-0.52	dvol_trend_21_126 z=-1.2; days_since_20pct_dd z=-0.9; streak_now z=-0.5	nan
HRL	Consumer Defensive	low	0.89	0.92	0.19	0.54	0.42	-0.46	-1.16	-0.59	0.16	1.15	dvol_trend_21_126 z=-0.9; nr7_count_20 z=-1.0; streak_now z=-0.4	nan
AVY	Consumer Cyclical	mid	0.89	0.88	0.66	0.4	-0.63	-0.12	-0.87	0.28	-1.3	-0.22	log_dvol_63 z=+1.7; rs_sec_63 z=+1.1; clv_21 z=-0.1	nan
EVERG	Utilities	low	0.89	0.91	0.91	-0.2	-0.46	-0.53	-1.08	0.29	0.3	1.55	days_since_20pct_dd z=+1.3; dvol_trend_21_126 z=+0.2; streak_now z=-0.7	nan
ABBV	Healthcare	low	0.89	0.9	0.97	-0.39	-0.3	1.19	-1.07	-1.59	0.54	-0.3	dvol_trend_21_126 z=-1.3; days_since_20pct_dd z=+0.4; streak_now z=+0.7	nan
KEYS	Technology	high	0.89	0.89	0.98	1.5	0.22	1.27	-0.51	-1.01	-0.96	-0	mom_12_1 z=+1.5; log_dvol_63 z=+1.8; days_since_52w_high z=-1.1	nan
DXCM	Healthcare	mid	0.88	0.87	0.88	2.36	0.41	1.43	-0.76	-0.79	0.13	-0.95	base_depth_126 z=+1.0; clv_21 z=+0.4; log_dvol_63 z=-0.0	nan
YUM	Consumer Cyclical	low	0.88	0.9	0.49	-0.85	-0.13	-2.83	0.23	0.49	1.09	0.56	days_since_20pct_dd z=+0.6; nr7_count_20 z=-0.9; dvol_trend_21_126 z=-0.5	nan
STX	Technology	high	0.88	0.88	0.9	0.5	-1.02	-1.37	-0.57	-0.29	0.81	0.71	amihud_21 z=-2.2; log_dvol_63 z=+2.1; dn_vol_63 z=+1.7	nan
CI	Healthcare	low	0.88	0.89	0.55	-0.25	0.37	0.11	-0.51	-0.21	-1.4	0.07	days_since_20pct_dd z=+0.9; nr7_count_20 z=-1.9; streak_now z=+0.7	nan
DHR	Healthcare	mid	0.88	0.87	0.77	0.11	0.59	0.77	-0.29	-0.09	0.22	0.15	rs_sec_63 z=+1.9; base_depth_126 z=+1.0; clv_21 z=+1.5	nan
C	Financial	high	0.87	0.87	0.6	0.25	-1.26	-0.97	-1	-1.01	0.4	1.32	amihud_21 z=-1.4; days_since_ignition z=+1.2; ignition_mult_10 z=-0.7	nan
PCAR	Industrials	high	0.87	0.87	0.26	0.32	-0.26	-1.46	-1.36	0.84	-0.51	0.58	amihud_21 z=-2.9; log_dvol_63 z=+1.5; up_day_frac_63 z=-1.9	nan
HAS	Consumer Cyclical	mid	0.87	0.86	0.56	0.38	-1.19	0.23	-0.45	-0.25	-0.51	-1.5	rs_sec_63 z=+1.6; base_depth_126 z=+0.6; sharpe_21 z=-0.2	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
WDC	Technology	high	0.87	0.86	0.76	0.25	-0.59	-2.5	-0.82	-0.58	0.91	0.99	amihud_21 z=-2.0; log_dvol_63 z=+1.7; dn_vol_63 z=+1.7	nan
FCX	Basic Materials	high	0.87	0.86	0.81	2.33	0.3	0.67	-0.72	-1.31	-1.4	-1.2	mom_12_1 z=+2.3; amihud_21 z=-1.9; days_since_52w_high z=-1.0	nan
PANW	Technology	high	0.86	0.85	0.97	0.81	0.22	1.28	0.2	-0.39	0.25	-1.05	log_dvol_63 z=+2.5; amihud_21 z=-1.7; mom_12_1 z=+0.8	nan
ATO	Utilities	low	0.86	0.89	0.99	-1.25	-0.95	-2.93	-1.5	-0.6	-0.1	1.49	dvol_trend_21_126 z=-1.0; streak_now z=+0.7; days_since_20pct_dd z=+0.0	nan
HST	Real Estate	mid	0.86	0.85	0.93	1.12	-0.71	-0.17	-1.26	0.05	1	1.4	log_dvol_63 z=+2.1; corr_spy_63 z=-2.1; sharpe_21 z=+0.6	nan
AME	Industrials	high	0.86	0.84	0.83	0.94	-0.23	0.72	-0.52	0.37	0.88	0.03	log_dvol_63 z=+1.9; mom_12_1 z=+0.9; amihud_21 z=-1.5	nan
NTRS	Financial	high	0.86	0.84	0.83	0.78	-0.21	-0.68	-0.13	0.05	1.13	1.26	log_dvol_63 z=+1.6; mom_12_1 z=+0.8; amihud_21 z=-1.6	nan
CAT	Industrials	high	0.85	0.83	0.58	0.33	-1.04	-1.56	-1.38	-0.92	0.76	1.3	amihud_21 z=-1.6; log_dvol_63 z=+1.5; days_since_2x_low z=-1.9	nan
AES	Utilities	mid	0.85	0.85	0.89	0.5	-1.74	0.93	-1.18	-0.64	0.93	-1.57	base_depth_126 z=+1.5; days_since_20pct_dd z=+2.9; log_dvol_63 z=-0.7	nan
SWK	Industrials	high	0.85	0.83	0.64	1.6	0.61	0.4	-0.77	-0.22	0.38	-0.34	mom_12_1 z=+1.6; amihud_21 z=-1.8; dn_vol_63 z=-0.7	nan
BEN	Financial	high	0.85	0.82	0.66	1.93	0.34	0.63	-1.46	-1.09	-0.87	0	mom_12_1 z=+1.9; amihud_21 z=-2.1; log_dvol_63 z=+1.5	nan
HUM	Healthcare	mid	0.85	0.84	0.84	2.93	0.5	1.57	-0.83	-1.12	0.1	-0.45	base_depth_126 z=+1.2; corr_spy_63 z=+1.0; days_since_20pct_dd z=+2.7	nan
IQV	Healthcare	mid	0.84	0.84	0.85	1.69	0.45	1.19	-1.17	-0.32	-0.35	-1.33	corr_spy_63 z=-2.5; base_depth_126 z=+1.1; rs_sec_63 z=+1.7	nan
CMI	Industrials	high	0.84	0.81	0.23	-0.29	-1.05	-2.95	-1.39	0.21	-0.01	1.45	amihud_21 z=-1.8; log_dvol_63 z=+1.6; up_day_frac_63 z=-2.1	nan
TROW	Financial	high	0.84	0.81	0.44	0.98	-0.41	-0.02	-0.76	-0.98	-1.59	1.51	mom_12_1 z=+1.0; days_since_52w_high z=-1.2; up_day_frac_63 z=-2.1	nan
FANG	Energy	low	0.84	0.88	0.37	1.1	-0.21	-0.1	0.04	1.17	-0.74	0.36	days_since_20pct_dd z=+1.0; dvol_trend_21_126 z=+0.9; streak_now z=-0.9	nan
GLW	Technology	high	0.84	0.8	0.72	0.11	-0.65	-2.59	-0.59	-1.11	0.82	1.09	log_dvol_63 z=+1.5; dn_vol_63 z=+1.9; days_since_ignition z=+1.4	nan
INTC	Technology	high	0.83	0.8	0.75	0.63	-0.36	0.51	-0.46	-0.55	-0.51	-0.36	amihud_21 z=-2.1; log_dvol_63 z=+2.1; mom_12_1 z=+0.6	nan
LH	Healthcare	mid	0.83	0.83	0.76	0.99	0.55	-0.65	0.09	0.91	-0.62	-0.34	log_dvol_63 z=+2.3; clv_21 z=+0.1; corr_spy_63 z=-0.7	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
PM	Consumer Defensive	low	0.83	0.87	0.81	-0.17	-0.02	-0.28	-1.28	-1.02	1.39	0.22	nr7_count_20 z=-1.5; dvol_trend_21_126 z=-0.6; streak_now z=+0.5	nan
AMAT	Technology	high	0.83	0.79	0.71	0.6	-0.91	-0.73	-0.53	-1.34	-0.22	-0.22	log_dvol_63 z=+1.9; amihud_21 z=-1.9; mom_12_1 z=+0.6	nan
SWKS	Technology	high	0.82	0.78	0.74	0.3	0.29	1.25	1.07	2.61	-2.88	-1.3	log_dvol_63 z=+2.7; days_since_52w_high z=-2.5; up_day_frac_63 z=+0.9	nan
TGT	Consumer Defensive	mid	0.82	0.82	0.85	3	0.49	0.83	-0.71	-1.14	-0.82	-0.69	days_since_20pct_dd z=+3.3; rs_sec_63 z=+1.6; corr_spy_63 z=-0.5	nan
BMJ	Healthcare	low	0.82	0.87	0.54	2.43	0.91	0.35	-0.8	-0.24	1.82	0.24	days_since_20pct_dd z=+2.7; dvol_trend_21_126 z=+0.8; streak_now z=-0.5	nan
TXN	Technology	high	0.82	0.78	0.8	1.4	-0.5	0.1	-0.85	-1.6	0.68	0	mom_12_1 z=+1.4; up_day_frac_63 z=+0.6; days_since_2x_low z=-2.1	nan
MO	Consumer Defensive	low	0.82	0.86	0.72	-0.66	-0.08	-1.17	-1.35	-0.69	0.23	1.28	dvol_trend_21_126 z=-0.8; days_since_20pct_dd z=+0.6; nr7_count_20 z=+0.0	nan
ADI	Technology	high	0.81	0.77	0.84	0.95	-0.52	0.41	-0.37	-1.35	0.3	-0.26	mom_12_1 z=+1.0; up_day_frac_63 z=+1.0; log_dvol_63 z=+1.4	nan
UNP	Industrials	low	0.81	0.86	0.92	1.25	0.19	-1.05	-0.45	-0.75	-1.53	0.42	dvol_trend_21_126 z=-1.0; days_since_20pct_dd z=+0.8; streak_now z=-0.5	nan
KLAC	Technology	high	0.81	0.77	0.77	-0.17	-0.82	-1.83	-0.78	-0.86	0.14	0.57	log_dvol_63 z=+1.6; amihud_21 z=-1.5; up_day_frac_63 z=-1.8	nan
ULTA	Consumer Cyclical	mid	0.81	0.82	0.55	-0.29	-1.5	-0.26	-0.79	-0.97	-0.92	-0.42	log_dvol_63 z=-1.7; base_depth_126 z=+0.8; rs_sec_63 z=+1.7	nan
MSI	Technology	low	0.81	0.85	0.82	-0.54	-0.56	0.14	-0.88	-1.45	-1.3	-0.95	dvol_trend_21_126 z=-1.0; nr7_count_20 z=+0.1; streak_now z=+0.2	nan
DE	Industrials	mid	0.8	0.81	0.81	2.47	0.67	0.87	-1.01	-0.35	-0.72	-1.02	rs_sec_63 z=+2.2; base_depth_126 z=+0.7; days_since_20pct_dd z=+1.8	nan
AKAM	Technology	high	0.8	0.76	0.4	0.87	-0.01	-1.43	-0.47	0.65	0.39	-0.44	mom_12_1 z=+0.9; log_dvol_63 z=+1.6; days_since_2x_low z=-1.8	nan
BRK-B	Financial	low	0.8	0.84	1	-1.04	0.01	-0.02	-0.65	-0.48	1.83	0.34	days_since_20pct_dd z=+0.6; streak_now z=+1.1; nr7_count_20 z=-0.9	nan
JKHY	Technology	low	0.8	0.84	0.22	1.29	-0.01	-1.47	0.22	-0.48	-0.26	-0.13	dvol_trend_21_126 z=-0.8; days_since_20pct_dd z=-1.0; nr7_count_20 z=-1.8	nan
ABNB	Consumer Cyclical	high	0.8	0.75	0.63	2.29	0.24	0.13	0.52	0.88	-0.99	-1.26	mom_12_1 z=+2.3; log_dvol_63 z=+1.9; days_since_52w_high z=-1.2	nan
DGX	Healthcare	low	0.79	0.83	0.7	1.54	0.6	-0.1	0.25	0.73	-1.27	-0.99	days_since_20pct_dd z=+1.0; dvol_trend_21_126 z=+1.2; streak_now z=-0.4	nan
BDX	Healthcare	mid	0.79	0.81	0.82	2.43	0.74	1.36	-0.19	-1.17	-0.26	-0.91	rs_sec_63 z=+1.9; log_dvol_63 z=-0.9; base_depth_126 z=+0.7	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
MMM	Industrials	mid	0.79	0.8	0.79	-0.6	-0.2	-0.45	-0.43	-0.47	-0.12	-0.26	corr_spy_63 z=-0.2; clv_21 z=+0.0; days_since_20pct_dd z=+1.1	nan
AEE	Utilities	low	0.79	0.83	0.86	-0.37	-0.29	-1.15	-1.02	0.43	0.71	1.83	days_since_20pct_dd z=+1.3; dvol_trend_21_126 z=+0.2; nr7_count_20 z=+0.8	nan
GS	Financial	high	0.79	0.75	0.31	-0.22	-0.7	-2.62	-0.78	0.05	-1.37	0.58	up_day_frac_63 z=-3.3; days_since_ignition z=+1.5; amihud_21 z=-1.4	nan
IFF	Basic Materials	mid	0.78	0.79	0.73	1.69	0.75	0.99	-1.26	-0.3	-0.79	-0.46	days_since_20pct_dd z=+2.1; base_depth_126 z=+0.7; rs_sec_63 z=+0.6	nan
AMGN	Healthcare	mid	0.78	0.79	0.65	2.21	0.73	-0.09	1.69	1.07	-0.67	0.47	log_dvol_63 z=+2.1; days_since_20pct_dd z=+1.7; rs_sec_63 z=+0.6	nan
ADP	Technology	low	0.78	0.82	0.63	-0.27	0.78	0.2	-0.59	-1.01	-0.64	-0.51	nr7_count_20 z=-1.8; dvol_trend_21_126 z=-0.4; days_since_20pct_dd z=-0.7	nan
DPZ	Consumer Cyclical	low	0.78	0.81	0.14	-1.04	-0.9	-1.66	-0	-0.7	-0.07	-0.51	dvol_trend_21_126 z=-1.4; days_since_20pct_dd z=-1.0; streak_now z=-0.4	nan
AIG	Financial	low	0.78	0.81	0.73	-0.98	-0.55	-1.25	-0.76	1	0.17	0.22	dvol_trend_21_126 z=+1.0; days_since_20pct_dd z=+0.8; nr7_count_20 z=-1.9	nan
LIN	Basic Materials	low	0.77	0.8	0.78	-0.73	0.07	-1.42	-0.69	-0.6	0.76	2.05	days_since_20pct_dd z=+0.6; streak_now z=+0.8; dvol_trend_21_126 z=-0.5	nan
FLEX	Technology	high	0.77	0.74	0.7	0.43	-0.99	-2.28	-0.76	-1.13	-0.94	0.95	days_since_52w_high z=+1.9; log_dvol_63 z=+1.4; dn_vol_63 z=+1.3	nan
PFG	Financial	mid	0.77	0.78	0.87	2.14	0.1	0.01	-0.75	-0.12	1.57	-1.34	log_dvol_63 z=+2.0; days_since_20pct_dd z=+1.9; sharpe_21 z=-1.7	nan
ALLE	Industrials	mid	0.77	0.78	0.73	-1.67	0.41	-0.51	-0.83	-0.18	-0.69	-1.19	corr_spy_63 z=-2.4; rs_sec_63 z=+1.8; log_dvol_63 z=+1.6	nan
A	Healthcare	high	0.77	0.73	0.84	0.49	0.36	1.42	0.32	0.52	-1.96	-1.01	log_dvol_63 z=+2.9; days_since_52w_high z=-1.5; dn_vol_63 z=-0.7	nan
PH	Industrials	mid	0.76	0.77	0.96	-0.62	-0.95	-1.05	-0.82	-0.82	0.9	0.31	sharpe_21 z=-0.2; rs_sec_63 z=+0.8; base_depth_126 z=-1.0	nan
CPT	Real Estate	low	0.76	0.8	0.59	0.05	0.26	-1.02	-1.07	-0.23	-0.43	1.76	days_since_20pct_dd z=+0.6; dvol_trend_21_126 z=-0.1; nr7_count_20 z=-0.1	nan
ROK	Industrials	high	0.76	0.73	0.8	0.4	-0.23	0.37	-0.94	-0.89	1.16	0.87	mom_12_1 z=+0.4; up_day_frac_63 z=+0.4; amihud_21 z=-1.2	nan
CIEN	Technology	high	0.76	0.72	0.59	-0.1	-0.41	-1.8	-0.89	-0.59	0.84	0.29	amihud_21 z=-1.5; log_dvol_63 z=+1.4; dn_vol_63 z=+1.1	nan
WELL	Real Estate	low	0.76	0.79	0.98	-0.11	-0.33	-2.45	-1.18	-1.27	-0.69	1.23	dvol_trend_21_126 z=-1.3; days_since_20pct_dd z=+0.8; streak_now z=-0.6	nan
P	Technology	high	0.75	0.72	0.99	-0.8	0.25	1.6	-0.61	3.77	0.91	-0.65	log_dvol_63 z=+2.0; days_since_52w_high z=-1.2; up_day_frac_63 z=+1.7	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
VRSK	Industrials	low	0.75	0.78	0.23	-1.06	0.39	-1.19	-0.45	-1.2	0.21	0	dvol_trend_21_126 z=-2.2; days_since_20pct_dd z=-1.2; streak_now z=-0.9	nan
EXC	Utilities	low	0.75	0.78	0.84	-0.52	-0.78	-2	-1.02	0.03	1.12	2.14	days_since_20pct_dd z=+1.3; streak_now z=+1.2; nr7_count_20 z=+0.2	nan
NWSA	Communication Services	mid	0.75	0.76	0.72	-0.28	0.37	0.15	-0.24	-0.16	-0.25	0.17	clv_21 z=+0.3; rs_sec_63 z=+1.0; up_day_frac_63 z=+0.6	nan
PGR	Financial	low	0.75	0.77	0.46	-0.8	0.39	-0.08	-0.54	-0.57	1.45	1.37	nr7_count_20 z=-1.1; days_since_20pct_dd z=-0.7; dvol_trend_21_126 z=-0.3	nan
BE	Industrials	high	0.74	0.71	0.82	-0.17	-0.59	0.59	-1.16	0.31	-0.24	0.05	amihud_21 z=-1.9; log_dvol_63 z=+1.7; days_since_2x_low z=+1.2	nan
DLR	Real Estate	mid	0.74	0.76	0.69	-0.71	-1.34	-0.64	0.26	-0.64	1.46	-0.77	rs_sec_63 z=+1.0; days_since_20pct_dd z=+1.4; clv_21 z=-1.8	nan
WEC	Utilities	low	0.74	0.77	0.69	-0.81	-0.68	-1.39	-0.39	1.35	0.04	1.7	days_since_20pct_dd z=+1.0; dvol_trend_21_126 z=+0.9; streak_now z=-0.8	nan
NUE	Basic Materials	high	0.74	0.7	0.92	2.14	0.01	0.55	0.31	-0.61	0.59	0.06	mom_12_1 z=+2.1; up_day_frac_63 z=+1.0; amihud_21 z=-0.4	nan
TDY	Technology	high	0.74	0.7	0.54	-0.72	-0.92	-0.88	-1.3	-1.14	2.75	0.42	amihud_21 z=-1.7; up_day_frac_63 z=+0.4; dn_vol_63 z=-0.3	nan
AAPL	Technology	mid	0.73	0.75	0.78	0.5	-0.73	0.75	-0.41	-0.59	-0.36	0.59	corr_spy_63 z=-2.6; base_depth_126 z=+0.7; log_dvol_63 z=+1.5	nan
VTR	Real Estate	low	0.73	0.76	0.46	0.5	0.27	-2.52	-0.8	0.46	0.21	2.08	days_since_20pct_dd z=+1.0; streak_now z=-0.6; dvol_trend_21_126 z=-0.5	nan
WM	Industrials	low	0.73	0.75	0.76	-0.75	-0.77	-2.43	-0.86	0.01	-0.48	0.72	days_since_20pct_dd z=+0.0; dvol_trend_21_126 z=-0.4; streak_now z=+0.6	nan
MNST	Consumer Defensive	low	0.73	0.75	0.58	0.66	0.47	-0.66	-0.45	-0.46	-0.26	0.8	days_since_20pct_dd z=+2.2; streak_now z=+1.2; nr7_count_20 z=-1.6	nan
ORLY	Consumer Cyclical	low	0.73	0.74	0.4	-2.46	-0.36	-1.95	-0.38	0.12	-0.52	-0.03	days_since_20pct_dd z=-2.7; dvol_trend_21_126 z=-0.9; streak_now z=-2.2	nan
CFG	Financial	high	0.72	0.69	0.61	0.3	-0.4	-0.41	-0.31	-0.18	-0.07	1.44	amihud_21 z=-1.5; up_day_frac_63 z=-1.4; jump_21 z=+1.6	nan
CMCSA	Communication Services	low	0.72	0.73	0.07	0.04	-0.14	-1.84	-0.2	-0.53	-0.44	0.22	days_since_20pct_dd z=-0.8; streak_now z=+0.0; dvol_trend_21_126 z=-0.4	nan
DIS	Communication Services	mid	0.72	0.75	0.58	-0.52	-0.1	0.56	-0.4	-0.56	-0.56	-0.04	log_dvol_63 z=-0.5; base_depth_126 z=+0.5; up_day_frac_63 z=+0.3	nan
ITW	Industrials	mid	0.72	0.74	0.61	-0.1	0.03	-0.97	-0.35	0.07	0.35	0.15	rs_sec_63 z=+1.4; clv_21 z=+0.1; corr_spy_63 z=-1.7	nan
LRCX	Technology	high	0.72	0.69	0.65	0.25	-1	-0.44	-0.55	-0.63	-0.37	-0.3	amihud_21 z=-2.1; log_dvol_63 z=+1.7; days_since_ignition z=-1.1	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
WTW	Financial	low	0.71	0.73	0.42	-0.75	0.71	-1.42	-0.56	-0.75	-2.58	-0.36	days_since_20pct_dd z=-1.1; streak_now z=-1.0; dvol_trend_21_126 z=-0.5	nan
HSIC	Healthcare	mid	0.71	0.73	0.76	3.27	0.64	0.92	-0.66	-0.6	0.05	0.71	days_since_20pct_dd z=+2.4; corr_spy_63 z=-0.0; base_depth_126 z=+0.4	nan
NWS	Communication Services	mid	0.71	0.73	0.62	-0.52	0.25	-0.23	-0.33	-0.47	-0.55	0.29	log_dvol_63 z=+1.4; clv_21 z=+0.2; up_day_frac_63 z=+0.7	nan
TER	Technology	high	0.71	0.68	0.86	0.6	-0.71	0.74	-0.65	-1.73	-0.94	-0.13	mom_12_1 z=+0.6; log_dvol_63 z=+1.5; up_day_frac_63 z=+0.2	nan
CINF	Financial	low	0.71	0.72	0.67	-0.59	-0.12	-1.46	-1.29	0.06	-0.12	2.37	days_since_20pct_dd z=+1.9; dvol_trend_21_126 z=-0.5; streak_now z=+0.3	nan
HSY	Consumer Defensive	low	0.7	0.72	0.43	-0.09	-0.98	-1.46	-1.34	-0.91	0.79	0.42	dvol_trend_21_126 z=-1.0; streak_now z=+1.3; days_since_20pct_dd z=-0.7	nan
CNC	Healthcare	mid	0.7	0.72	0.67	2.03	0.64	0.91	-0.48	-0.64	-0.88	-0.13	corr_spy_63 z=+1.6; base_depth_126 z=+0.6; log_dvol_63 z=+0.1	nan
CRWD	Technology	high	0.7	0.67	0.99	0.26	0.26	1.15	-0.1	-0.22	-1.47	-1.24	amihud_21 z=-2.1; days_since_52w_high z=-0.9; days_since_ignition z=-0.9	nan
ROST	Consumer Cyclical	mid	0.7	0.72	0.82	1.05	-0.26	-0.85	-0.89	-1.24	0.92	0.52	up_day_frac_63 z=+0.7; log_dvol_63 z=+1.3; base_depth_126 z=-1.0	nan
CF	Basic Materials	low	0.7	0.71	0.33	2.21	-0.5	-0.62	-0.5	-0.61	-0.09	-0.38	days_since_2x_low z=-1.1; nr7_count_20 z=-1.6; streak_now z=+0.7	nan
CAH	Healthcare	low	0.69	0.7	0.68	1.05	0.53	-0.72	-0.25	-0.31	-1.01	0.31	days_since_20pct_dd z=-2.1; dvol_trend_21_126 z=-0.7; nr7_count_20 z=-1.0	nan
DUK	Utilities	low	0.69	0.7	0.93	-0.88	-0.95	-2.37	-1.46	0.32	-0.24	2.43	days_since_20pct_dd z=+0.6; nr7_count_20 z=+1.8; streak_now z=-0.5	nan
LMT	Industrials	low	0.69	0.69	0.32	0.02	-1.03	-1.27	-1.09	-0.77	0.66	-0.56	days_since_20pct_dd z=-0.9; dvol_trend_21_126 z=-0.5; nr7_count_20 z=-0.8	nan
GILD	Healthcare	mid	0.69	0.71	0.75	0.7	-0.12	0.47	-0.9	-0.9	-1.14	-0.05	corr_spy_63 z=-0.5; clv_21 z=-0.0; days_since_20pct_dd z=-0.9	nan
VLO	Energy	low	0.69	0.69	0.44	2.57	0.72	1.3	0.07	1.58	-0.13	-1.73	days_since_20pct_dd z=+2.2; days_since_2x_low z=-1.4; dvol_trend_21_126 z=+3.0	nan
VLTO	Industrials	mid	0.68	0.7	0.8	-0.68	-0.26	0.48	-0.94	-0.48	0.91	-0.33	rs_sec_63 z=+1.5; up_day_frac_63 z=+0.9; days_since_20pct_dd z=-1.0	nan
SRE	Utilities	low	0.68	0.68	0.41	-1.01	-1.3	-1.74	-0.51	-0.47	-0.12	0.95	days_since_20pct_dd z=-1.5; dvol_trend_21_126 z=-1.0; streak_now z=-0.5	nan
GOOGL	Communication Services	high	0.68	0.67	0.53	-0.16	-0.76	-0.91	-0.52	-0.7	1.04	0.59	days_since_52w_high z=+1.8; dn_vol_63 z=+1.1; days_since_ignition z=+1.3	nan
XEL	Utilities	low	0.68	0.67	0.81	-0.76	-0.89	-0.81	-0.24	0.37	-0.44	1.06	days_since_20pct_dd z=+1.4; streak_now z=+1.2; nr7_count_20 z=+0.8	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
TPR	Consumer Cyclical	high	0.68	0.66	0.43	-1	-0.55	-1.64	-0.76	0.75	-0.56	1.68	dn_vol_63 z=+1.8; mom_12_1 z=-1.0; ignition_mult_10 z=+0.5	nan
TRV	Financial	low	0.67	0.67	0.89	1.05	-0.05	-1.14	-0.46	-1.2	-0.13	-0.72	dvol_trend_21_126 z=-1.3; days_since_20pct_dd z=+0.0; streak_now z=+0.3	nan
SW	Consumer Cyclical	high	0.67	0.66	0.3	1.19	-0.19	0.29	-1.31	-1.5	nan	0.02	mom_12_1 z=+1.2; amihud_21 z=-0.0; days_since_52w_high z=-3.4	nan
WDAY	Technology	mid	0.67	0.7	0.44	-0.39	0.53	-0.01	-1.07	-1.48	-0.3	-1.09	rs_sec_63 z=+1.2; base_depth_126 z=+0.2; corr_spy_63 z=-0.8	nan
AIZ	Financial	low	0.67	0.66	0.66	0.69	0.45	-1.21	-0.51	0.37	-0.4	-0.25	days_since_20pct_dd z=+1.5; streak_now z=-0.6; dvol_trend_21_126 z=+0.0	nan
NXPI	Technology	high	0.67	0.65	0.33	-0.61	-0.39	-1.13	-0.49	-0.16	-0.08	0.87	amihud_21 z=-1.4; log_dvol_63 z=+1.3; days_since_ignition z=+1.0	nan
BR	Technology	low	0.66	0.66	0.33	-1.34	0.93	-1.28	-0.72	-0.9	-0.81	-0.77	days_since_20pct_dd z=-1.4; dvol_trend_21_126 z=-1.1; nr7_count_20 z=-0.6	nan
MAR	Consumer Cyclical	mid	0.66	0.69	0.74	0.13	-0.67	-0.81	-0.9	0.31	1.32	1.4	corr_spy_63 z=-2.7; sharpe_21 z=+0.8; log_dvol_63 z=+1.5	nan
JBL	Technology	high	0.66	0.64	0.61	-0.22	-0.35	-1.65	-0.08	0.75	0.99	0.57	dn_vol_63 z=+1.2; log_dvol_63 z=+1.3; days_since_ignition z=+1.2	nan
CPRT	Industrials	low	0.66	0.65	0.08	-0.92	-1.08	-1.23	0.21	-0.18	-1.5	-1.09	days_since_20pct_dd z=-1.0; dvol_trend_21_126 z=-0.6; nr7_count_20 z=-0.9	nan
IT	Technology	mid	0.66	0.68	0.56	-0.44	0.26	0.03	-0.2	-0.97	-1.58	-1.08	base_depth_126 z=+0.8; rs_sec_63 z=+1.5; log_dvol_63 z=-0.0	nan
PNW	Utilities	low	0.65	0.64	0.74	0.26	-0.02	-1.3	-0.05	-0.02	0.43	1.91	days_since_20pct_dd z=+0.6; streak_now z=+1.1; nr7_count_20 z=-0.0	nan
ODFL	Industrials	high	0.65	0.64	0.23	0.98	-0.07	-0.9	-1.14	-0.76	-0.25	0.72	mom_12_1 z=+1.0; amihud_21 z=-1.6; up_day_frac_63 z=-1.4	nan
AJG	Financial	low	0.65	0.64	0.28	-0.96	0.4	-1.06	-0.39	-0.89	-1	-0.4	days_since_20pct_dd z=-1.5; nr7_count_20 z=-1.0; dvol_trend_21_126 z=-0.5	nan
PRU	Financial	mid	0.65	0.68	0.79	0.9	-0.3	-0.07	-0.63	-1.07	0.62	0.01	sharpe_21 z=-2.0; corr_spy_63 z=-0.7; rs_sec_63 z=+0.6	nan
FFIV	Technology	high	0.65	0.63	0.93	-0.4	0.06	1.06	-0.61	-1.15	-0.01	0.06	days_since_2x_low z=-6.0; days_since_52w_high z=-0.7; log_dvol_63 z=+1.2	nan
GOOG	Communication Services	high	0.64	0.63	0.54	-0.2	-0.79	-0.89	-0.53	-0.64	0.87	0.53	days_since_52w_high z=+1.8; days_since_ignition z=+1.2; dn_vol_63 z=+0.9	nan
BAX	Healthcare	high	0.64	0.62	0.48	1.66	0.85	0.86	-0.97	-0.46	-1.52	-1.18	mom_12_1 z=+1.7; days_since_52w_high z=-3.5; dn_vol_63 z=-0.5	nan
MS	Financial	high	0.64	0.61	0.5	0.54	-0.9	-1.57	-0.76	-0.18	-0.89	1.12	amihud_21 z=-1.7; mom_12_1 z=+0.5; up_day_frac_63 z=-2.1	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
MCK	Healthcare	low	0.64	0.63	0.55	-0.52	-0.53	-0.13	-0.25	-0.13	-1.12	-0.86	dvol_trend_21_126 z=-0.8; days_since_20pct_dd z=-1.0; streak_now z=+1.0	nan
CNP	Utilities	low	0.64	0.63	0.64	-0.57	-0.65	-2.38	-0.38	0.96	-1.23	1.78	days_since_20pct_dd z=+0.6; streak_now z=-0.6; nr7_count_20 z=-1.0	nan
WAB	Industrials	high	0.63	0.61	0.73	0.47	-0.7	0.11	-0.93	-0.61	0.71	-0.67	mom_12_1 z=+0.5; ignition_mult_10 z=+0.8; log_dvol_63 z=+1.2	nan
AMD	Technology	high	0.63	0.6	0.96	1.36	0.21	1.35	-0.91	-1.14	1.12	-0.33	mom_12_1 z=+1.4; days_since_52w_high z=-1.0; up_day_frac_63 z=+0.7	nan
CPAY	Technology	mid	0.63	0.67	0.78	1.49	0.53	0.69	-1.25	-1.12	-0.45	-0.79	base_depth_126 z=+0.6; log_dvol_63 z=+1.4; corr_spy_63 z=-1.7	nan
CARR	Industrials	high	0.63	0.6	0.16	-0.51	-0.48	-1.19	-1.14	0.08	0.07	1.54	amihud_21 z=-1.3; dn_vol_63 z=+0.8; up_day_frac_63 z=-1.3	nan
ACN	Technology	mid	0.63	0.67	0.29	-0.59	-0.71	-0.57	1.24	-0.2	-0.88	-1.63	rs_sec_63 z=+1.9; clv_21 z=-2.6; sharpe_21 z=+0.1	nan
V	Financial	mid	0.62	0.66	0.88	-0.38	0.32	0.44	-0.63	-1.09	-0.9	-0.66	rs_sec_63 z=+1.8; days_since_20pct_dd z=-1.5; base_depth_126 z=+0.4	nan
ES	Utilities	low	0.62	0.62	0.83	-0.03	-0.12	-0.27	-0.59	0.06	0.09	1.13	days_since_20pct_dd z=+1.6; streak_now z=+2.1; dvol_trend_21_126 z=+0.2	nan
LYB	Basic Materials	low	0.62	0.61	0.3	1.7	-1.13	-0.46	-0.78	-1.64	0.74	-0.63	days_since_2x_low z=-1.3; dvol_trend_21_126 z=-1.3; streak_now z=+1.3	nan
REGN	Healthcare	mid	0.62	0.65	0.42	1.26	-0.13	0.26	-0.6	-0.62	0.81	-0.5	corr_spy_63 z=-0.1; log_dvol_63 z=-0.2; sharpe_21 z=-1.9	nan
WMT	Consumer Defensive	low	0.62	0.61	0.25	-1.39	-0.71	-2.82	-0.86	-0.39	0.45	1.46	days_since_20pct_dd z=-1.8; dvol_trend_21_126 z=-0.6; streak_now z=+0.3	nan
MDLZ	Consumer Defensive	low	0.61	0.6	0.77	0.35	0.46	0.13	-1.59	-0.88	0.73	-0.69	days_since_20pct_dd z=+0.4; streak_now z=+0.9; nr7_count_20 z=+0.5	nan
MKC	Consumer Defensive	low	0.61	0.6	0.11	-0.81	0.15	-1.65	-0.33	-0.27	-0.78	0.8	streak_now z=+0.9; nr7_count_20 z=-0.7; days_since_20pct_dd z=-0.7	nan
SO	Utilities	low	0.61	0.59	0.8	-1.45	-0.9	-2.42	-0.84	0.47	0.91	2.19	days_since_20pct_dd z=+0.7; streak_now z=+1.2; dvol_trend_21_126 z=+0.1	nan
OKE	Energy	low	0.61	0.58	0.65	0.91	-0.26	0.16	-0.52	-0.46	0.8	-0.24	streak_now z=+1.0; nr7_count_20 z=-0.8; dvol_trend_21_126 z=-0.1	nan
EW	Healthcare	mid	0.61	0.65	0.65	1.15	0.1	0.52	-0	-0.91	0.66	0.31	days_since_20pct_dd z=+2.5; corr_spy_63 z=-1.8; log_dvol_63 z=+0.0	nan
MOS	Basic Materials	mid	0.6	0.64	0.13	-0.62	-1.36	-0.93	-0.75	-0.45	-0.1	-1.11	rs_sec_63 z=+0.6; sharpe_21 z=-1.7; base_depth_126 z=-0.2	nan
COHR	Technology	high	0.6	0.59	0.77	0.34	-1.17	-0.4	-0.32	0.07	0.38	0.39	log_dvol_63 z=+1.6; amihud_21 z=-1.6; dn_vol_63 z=+1.4	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
TRGP	Energy	low	0.6	0.58	0.63	0.99	0.24	0.27	-0.74	-0.63	0.04	-0.25	days_since_2x_low z=-2.1; streak_now z=+1.0; dvol_trend_21_126 z=-0.4	nan
MPWR	Technology	high	0.6	0.58	0.67	-0.28	-1.44	0.22	-0.19	-0.08	0.14	-0.16	log_dvol_63 z=+1.7; amihud_21 z=-1.6; days_since_ignition z=-1.0	nan
BRO	Financial	low	0.6	0.57	0.17	-1	-0.08	-1.15	-0.57	-0.92	-1.26	-0.02	days_since_20pct_dd z=-1.1; dvol_trend_21_126 z=-0.7; nr7_count_20 z=-0.0	nan
AOS	Industrials	mid	0.59	0.64	0.27	-1.04	-1.77	-1.42	-1.07	-0.29	-0.39	-0.16	corr_spy_63 z=-0.1; days_since_20pct_dd z=-0.8; rs_sec_63 z=+0.6	nan
IRM	Real Estate	high	0.59	0.58	0.57	-0.54	-1	-0.38	-0.95	-0.59	1.91	0.54	log_dvol_63 z=+1.3; jump_21 z=-1.6; days_since_2x_low z=+0.7	nan
NVR	Consumer Cyclical	mid	0.59	0.63	0.22	-1.26	-0.98	-1.17	-0.68	2.32	-1.88	0.52	log_dvol_63 z=+2.4; corr_spy_63 z=-0.6; days_since_20pct_dd z=-0.8	nan
DAL	Industrials	high	0.59	0.57	0.66	0.55	-0.71	0.17	-1.02	-0.38	0.51	0.53	mom_12_1 z=+0.6; log_dvol_63 z=+0.5; days_since_ignition z=-1.2	nan
MSCI	Financial	mid	0.59	0.62	0.44	-0.23	0.11	-0.72	-0.43	-0.46	-0.4	1.15	days_since_20pct_dd z=+2.0; clv_21 z=+0.1; base_depth_126 z=-0.9	nan
CSX	Industrials	mid	0.58	0.62	0.64	1.43	-0.55	-0.27	-0.64	-0.73	0.55	-0.13	corr_spy_63 z=-2.6; rs_sec_63 z=+0.7; days_since_20pct_dd z=+0.9	nan
SOLV	Healthcare	mid	0.58	0.61	0.75	0.96	0.29	1.11	-1.55	-1.18	-0.27	-0.82	base_depth_126 z=+0.7; log_dvol_63 z=+0.1; rs_sec_63 z=+0.8	nan
DVN	Energy	low	0.58	0.57	0.27	1.84	-0.89	1.07	-0.13	0.1	-0.17	-0.56	days_since_2x_low z=-1.2; days_since_20pct_dd z=+0.7; streak_now z=+1.2	nan
DRI	Consumer Cyclical	low	0.58	0.56	0.35	0.2	0.91	-0.72	0.13	-0.01	0.48	0.32	days_since_20pct_dd z=+0.9; dvol_trend_21_126 z=+0.3; nr7_count_20 z=-1.1	nan
DOW	Basic Materials	low	0.58	0.55	0.29	1.4	-1.08	-0.41	-1.1	-1.28	0.7	-0.34	days_since_2x_low z=-1.4; dvol_trend_21_126 z=-1.4; streak_now z=+1.4	nan
CBOE	Financial	low	0.57	0.55	0.09	-0.27	0.15	-2.43	0.53	0.27	0.51	0.13	days_since_20pct_dd z=-2.7; dvol_trend_21_126 z=-1.0; streak_now z=+0.3	nan
MRK	Healthcare	low	0.57	0.54	0.5	2.23	0.74	0.32	-0.93	-0.66	0.54	-1.02	days_since_2x_low z=-4.7; dvol_trend_21_126 z=+0.3; nr7_count_20 z=+1.0	nan
AFL	Financial	low	0.57	0.54	0.93	-0.84	0.13	-2.37	-0.77	-0.75	0.66	0.73	dvol_trend_21_126 z=-1.2; streak_now z=+0.7; days_since_20pct_dd z=+0.0	nan
BALL	Consumer Cyclical	low	0.57	0.53	0.61	1.13	-0.43	-0.17	-0.54	-0.95	-0.49	0.39	dvol_trend_21_126 z=-1.1; days_since_20pct_dd z=+0.2; nr7_count_20 z=-0.0	nan
GWV	Industrials	mid	0.57	0.61	0.64	0.89	-0.42	0.03	-0.55	-0.04	-0.27	0.78	corr_spy_63 z=-3.2; log_dvol_63 z=+1.5; sharpe_21 z=-0.6	nan
MAS	Industrials	high	0.56	0.57	0.43	-0.48	-0.32	-0.64	-0.87	-0.56	0.47	1.79	dn_vol_63 z=+2.1; amihud_21 z=-0.5; days_since_52w_high z=-0.4	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
WRB	Financial	low	0.56	0.52	0.73	-1.51	-0.11	-0.53	-1.02	2.2	-0.11	0.19	dvol_trend_21_126 z=+2.7; days_since_20pct_dd z=+0.9; streak_now z=+1.2	nan
NOW	Technology	mid	0.56	0.6	0.48	-0.95	0.26	-0.48	-0.84	-1.31	-0.33	-0.94	base_depth_126 z=+0.5; rs_sec_63 z=+0.8; days_since_20pct_dd z=-0.8	nan
PWR	Industrials	high	0.56	0.56	0.57	-0.03	-1.45	-0.58	-0.72	-1.04	1.03	-0.53	days_since_52w_high z=+2.9; log_dvol_63 z=+1.3; ignition_mult_10 z=-0.7	nan
NEM	Basic Materials	high	0.56	0.55	0.64	0.2	0.28	0.11	-1.55	-0.91	-1.54	-1.11	dn_vol_63 z=-0.5; days_since_52w_high z=-0.7; amihud_21 z=-1.2	nan
PG	Consumer Defensive	low	0.55	0.52	0.95	-0.38	-0.1	-0.62	-0.66	-1.52	1.34	0.47	dvol_trend_21_126 z=-0.8; days_since_20pct_dd z=-1.0; nr7_count_20 z=+0.7	nan
PPG	Basic Materials	high	0.55	0.55	0.39	0.99	0.31	-0.81	-1.12	-0.8	0.45	1.1	log_dvol_63 z=-1.0; mom_12_1 z=+1.0; amihud_21 z=-0.4	nan
RMD	Healthcare	mid	0.55	0.59	0.41	-0.83	-1.42	-0.46	-0.69	-0.29	-0.85	0.05	corr_spy_63 z=-1.7; base_depth_126 z=+0.5; days_since_20pct_dd z=-0.9	nan
PSX	Energy	low	0.55	0.51	0.54	2.4	0.42	0.98	-0.36	1.13	-2.29	-2.14	days_since_2x_low z=-2.3; days_since_20pct_dd z=+1.3; dvol_trend_21_126 z=+2.2	nan
DG	Consumer Defensive	mid	0.55	0.59	0.39	0.91	-0.43	0.41	-0.54	-0.75	-0.27	-0.3	log_dvol_63 z=-1.0; corr_spy_63 z=+0.1; base_depth_126 z=+0.6	nan
CHD	Consumer Defensive	low	0.54	0.51	0.87	0.67	0.41	0.17	-1.69	-0.83	-1.03	-0.05	nr7_count_20 z=+0.8; streak_now z=+0.7; dvol_trend_21_126 z=-0.4	nan
KVUE	Consumer Defensive	low	0.54	0.5	0.72	1.53	0.25	0.48	-0.38	-0.45	-0.31	0.85	dvol_trend_21_126 z=+0.9; days_since_20pct_dd z=+0.1; streak_now z=+0.4	nan
IR	Industrials	high	0.54	0.54	0.2	-0.93	-1.36	-0.82	-0.8	1.3	-0.1	-0.77	log_dvol_63 z=+2.3; amihud_21 z=-1.7; days_since_ignition z=-1.7	nan
KDP	Consumer Defensive	low	0.54	0.49	0.6	2.63	0.25	0.69	-0.36	-0.64	-1.48	-0.61	nr7_count_20 z=-1.9; dvol_trend_21_126 z=-0.5; days_since_2x_low z=+0.0	nan
CRM	Technology	mid	0.54	0.58	0.33	0.06	0.48	0.57	-0.73	-0.52	-1.15	-1.66	clv_21 z=-3.0; rs_sec_63 z=+1.5; log_dvol_63 z=+1.4	nan
GPC	Consumer Cyclical	mid	0.53	0.58	0.5	0.63	0.91	0.35	-0.76	-1.64	-0.77	-0.25	base_depth_126 z=+0.5; log_dvol_63 z=+0.4; sharpe_21 z=-1.4	nan
SNDK	Technology	high	0.53	0.54	0.62	nan	-1.1	nan	nan	nan	nan	0.55	days_since_52w_high z=-1.5; ignition_mult_10 z=-0.7; days_since_ignition z=-0.3	nan
APA	Energy	low	0.53	0.49	0.23	1.68	0.06	1.31	-0.38	-0.47	-0.68	-1.33	days_since_2x_low z=-1.9; dvol_trend_21_126 z=+0.6; streak_now z=+1.2	nan
META	Communication Services	high	0.53	0.53	0.72	-1.24	0.14	0.7	0.36	1.61	-0.43	-0.32	log_dvol_63 z=+1.6; mom_12_1 z=-1.2; days_since_ignition z=-1.6	nan
0	Real Estate	low	0.53	0.48	0.61	0.18	-0.22	-1.55	-0.7	1.8	-0.78	0.48	days_since_20pct_dd z=+1.1; dvol_trend_21_126 z=+2.2; nr7_count_20 z=-1.5	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
CVX	Energy	low	0.52	0.48	0.67	2.35	0.21	0.66	-0.5	-0.2	-1.25	-1.43	dvol_trend_21_126 z=+0.5; streak_now z=-1.0; nr7_count_20 z=+0.8	nan
LUV	Industrials	high	0.52	0.52	0.46	0.43	-0.38	-0.48	-1.33	-0.91	1.78	0.81	log_dvol_63 z=-0.7; amihud_21 z=-0.3; mom_12_1 z=+0.4	nan
NEE	Utilities	low	0.52	0.47	0.48	0.05	-0.98	-0.85	-0.63	0.59	-0.4	1.62	days_since_20pct_dd z=-1.1; streak_now z=-0.6; dvol_trend_21_126 z=-0.0	nan
ZBH	Healthcare	low	0.52	0.46	0.42	1.22	0.24	0.99	0.31	-0.21	-0.28	-0.62	days_since_20pct_dd z=+1.8; streak_now z=+0.8; dvol_trend_21_126 z=-0.1	nan
FTV	Technology	mid	0.52	0.57	0.6	0.81	-0.46	0.17	-0.42	-1.29	0.3	0.61	log_dvol_63 z=-0.8; rs_sec_63 z=-1.1; sharpe_21 z=-0.1	nan
MA	Financial	mid	0.51	0.56	0.81	-0.9	0.39	0.03	-0.98	-1.34	-1.38	-1.22	rs_sec_63 z=+2.6; days_since_20pct_dd z=-2.1; corr_spy_63 z=-1.6	nan
AMP	Financial	mid	0.51	0.56	0.7	-0.02	0.31	-0.75	-0.15	0.35	0.4	0.91	sharpe_21 z=-2.1; up_day_frac_63 z=+0.4; base_depth_126 z=-0.9	nan
QCOM	Technology	high	0.51	0.52	0.49	-0.52	-0.33	-0.8	0.44	-0.08	-0.43	-0.24	days_since_2x_low z=-0.9; ignition_mult_10 z=-0.4; days_since_52w_high z=-0.2	nan
JNJ	Healthcare	low	0.51	0.46	0.94	1.64	1.15	-0.69	-0.79	-0.3	1.7	0.92	streak_now z=-2.0; nr7_count_20 z=-0.8; dvol_trend_21_126 z=-0.1	nan
SCHW	Financial	mid	0.51	0.55	0.58	0.21	0.15	-0.43	0.45	-0.28	-0.08	1.05	days_since_20pct_dd z=+2.0; sharpe_21 z=-1.7; base_depth_126 z=-0.9	nan
EBAY	Consumer Cyclical	mid	0.5	0.55	0.53	-0.5	-0.71	-0.28	-0.8	0.16	-0.47	0.48	corr_spy_63 z=-1.8; sharpe_21 z=-0.2; clv_21 z=-0.3	nan
USB	Financial	mid	0.5	0.54	0.68	1.01	-0.22	-0.33	-0.29	-0.16	0.34	1.76	corr_spy_63 z=-0.0; days_since_20pct_dd z=+1.9; sharpe_21 z=-2.0	nan
JPM	Financial	mid	0.5	0.53	0.9	-1	-0.7	-1.04	-0.69	-0.27	2.61	1.01	corr_spy_63 z=-0.0; sharpe_21 z=-2.0; base_depth_126 z=-1.0	nan
TRMB	Technology	high	0.5	0.51	0.19	-1.51	-1.16	-1.35	-0.93	-0.26	-3.01	-0.68	mom_12_1 z=-1.5; log_dvol_63 z=+1.5; amihud_21 z=-1.4	nan
IBKR	Financial	high	0.49	0.51	0.56	-0.76	-0.68	-0.28	-0.94	-0.56	0.41	0.62	up_day_frac_63 z=-2.4; jump_21 z=-1.7; mom_12_1 z=-0.8	nan
DVA	Healthcare	low	0.49	0.45	0.12	0.45	-0.3	-1.16	0.1	-0.8	1.28	1.9	days_since_2x_low z=-0.9; dvol_trend_21_126 z=-1.1; streak_now z=+1.1	nan
NSC	Industrials	mid	0.49	0.53	0.52	-0.07	-0.42	-0.55	-0.68	-0.48	-0.36	-0.01	corr_spy_63 z=-2.2; days_since_20pct_dd z=+1.8; rs_sec_63 z=+0.8	nan
ESS	Real Estate	low	0.49	0.45	0.89	-0.1	-0.16	-0.17	-1.19	0.42	0.72	1.84	days_since_20pct_dd z=+0.9; dvol_trend_21_126 z=+0.5; nr7_count_20 z=-0.9	nan
CB	Financial	low	0.49	0.44	0.9	0.61	0.08	-1.33	-1.02	-0.32	1.09	1.24	streak_now z=-1.0; nr7_count_20 z=+0.1; days_since_20pct_dd z=+0.0	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
ELV	Healthcare	mid	0.48	0.52	0.5	1.39	0.28	0.69	-0.55	-0.96	-0.83	0.18	log_dvol_63 z=-1.1; corr_spy_63 z=-0.1; base_depth_126 z=+0.4	nan
CDNS	Technology	high	0.48	0.5	0.48	-1.46	-0.11	-0.42	-0.44	0.54	1.38	0.7	mom_12_1 z=-1.5; log_dvol_63 z=+1.4; dn_vol_63 z=+1.1	nan
GIS	Consumer Defensive	low	0.48	0.43	0.05	-0.18	1.37	-1.22	0.03	0.44	1.29	0.31	dvol_trend_21_126 z=+0.4; nr7_count_20 z=+0.9; streak_now z=-0.4	nan
EXR	Real Estate	mid	0.48	0.52	0.39	0	-0.08	-0.29	-0.56	1.67	0.82	0.2	days_since_20pct_dd z=+1.7; up_day_frac_63 z=-1.5; clv_21 z=-1.4	nan
UNH	Healthcare	mid	0.48	0.51	0.47	0.74	0.3	0.23	-0.7	-0.66	0.74	0.08	corr_spy_63 z=+0.0; log_dvol_63 z=-0.5; up_day_frac_63 z=-1.5	nan
NDAQ	Financial	mid	0.47	0.5	0.57	-0.22	-0.04	-0.1	-0.34	-0.49	-2.13	-0.77	rs_sec_63 z=+1.4; up_day_frac_63 z=-2.1; log_dvol_63 z=+1.1	nan
NOC	Industrials	low	0.47	0.43	0.15	-1.42	-0.75	-2.91	-0.67	0.48	0.44	0.92	days_since_20pct_dd z=-1.4; streak_now z=-2.7; nr7_count_20 z=+0.7	nan
CTSH	Technology	low	0.47	0.42	0.07	-0.27	-1.04	-1.29	0.25	-0.55	-0.85	-1.69	dvol_trend_21_126 z=-0.9; streak_now z=-0.9; days_since_20pct_dd z=-0.9	nan
WMB	Energy	low	0.47	0.42	0.75	-0.86	-0.8	-1.56	-0.64	0.38	1.3	0.5	days_since_20pct_dd z=+0.8; nr7_count_20 z=-1.1; streak_now z=+1.1	nan
INCY	Healthcare	mid	0.47	0.5	0.3	1.24	-0.27	0.1	-1.01	-0.31	-2.38	-0.42	days_since_20pct_dd z=+2.4; log_dvol_63 z=+1.4; up_day_frac_63 z=-1.9	nan
GEV	Industrials	high	0.46	0.49	0.75	-1.32	-1.31	-0.86	-0.35	-1.17	2.14	1.39	mom_12_1 z=-1.3; days_since_52w_high z=-0.7; days_since_2x_low z=-1.7	nan
DLTR	Consumer Defensive	mid	0.46	0.49	0.37	1.21	0.62	0.36	-0.39	-0.54	1.79	0.31	up_day_frac_63 z=+1.1; log_dvol_63 z=-0.2; clv_21 z=-1.9	nan
JBHT	Industrials	mid	0.46	0.48	0.32	2.46	-0.1	-0.69	1.08	0.32	0.39	1.09	log_dvol_63 z=+2.0; base_depth_126 z=-1.1; clv_21 z=-0.1	nan
MCHP	Technology	high	0.46	0.49	0.34	0.3	-0.42	-0.11	-0.73	-1.09	-0.02	-0.58	mom_12_1 z=+0.3; days_since_ignition z=+0.1; ignition_mult_10 z=+0.5	nan
PKG	Consumer Cyclical	mid	0.46	0.48	0.68	-0.58	-0.28	-0.25	-0.87	-1.2	-0.86	-0.45	up_day_frac_63 z=-1.7; log_dvol_63 z=+0.2; corr_spy_63 z=-0.6	nan
ADBE	Technology	mid	0.45	0.47	0.32	-0.4	0.81	-0.54	-0.56	-0.22	1.59	-0.88	log_dvol_63 z=-1.8; up_day_frac_63 z=+1.3; corr_spy_63 z=-1.6	nan
MTB	Financial	mid	0.45	0.47	0.45	0.31	-0.52	-0.47	-0.46	0.16	0.26	1.23	days_since_20pct_dd z=+1.9; log_dvol_63 z=+1.4; corr_spy_63 z=-0.3	nan
AMCR	Consumer Cyclical	mid	0.45	0.46	0.43	1.28	1	-0.02	-1.21	-0.98	-0.32	0.02	corr_spy_63 z=-0.3; log_dvol_63 z=+0.5; sharpe_21 z=-1.5	nan
CVS	Healthcare	low	0.45	0.41	0.21	0.71	0.31	-0.45	-0.42	0.21	-0.97	0.29	days_since_20pct_dd z=-0.9; nr7_count_20 z=+0.4; streak_now z=+0.7	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
URI	Industrials	high	0.45	0.48	0.69	-1.14	-0.88	0.3	-0.43	-0.33	0.44	0.07	mom_12_1 z=-1.1; days_since_ignition z=+1.3; up_day_frac_63 z=+0.5	nan
PODD	Healthcare	mid	0.44	0.45	0.1	-1.29	-1.57	-1.63	-0.65	0.95	1.41	1.65	clv_21 z=-0.1; rs_sec_63 z=-0.7; days_since_20pct_dd z=-0.7	nan
CME	Financial	low	0.44	0.4	0.45	-0.72	-1.2	-1.59	-0.23	-1.45	-0.95	-1.01	days_since_20pct_dd z=-1.8; dvol_trend_21_126 z=-2.0; nr7_count_20 z=-0.9	nan
SHW	Basic Materials	mid	0.44	0.45	0.4	-0.83	-0.04	-0.99	-0.97	-0.88	-0.28	-0.29	corr_spy_63 z=-0.3; clv_21 z=+0.2; days_since_20pct_dd z=-1.0	nan
TSCO	Consumer Cyclical	mid	0.44	0.44	0.25	-1.79	-1.46	-1.92	-0.41	0.27	-0.24	-0.69	base_depth_126 z=-1.3; rs_sec_63 z=+0.9; days_since_20pct_dd z=-1.0	nan
AZO	Consumer Cyclical	low	0.44	0.4	0.11	-2.71	-0.31	-2.37	-0.65	1.6	-0.89	0.35	days_since_20pct_dd z=-1.6; dvol_trend_21_126 z=+0.4; nr7_count_20 z=-1.8	nan
F	Consumer Cyclical	high	0.43	0.48	0.28	0.66	-0.1	-1.28	-0.34	-0.69	1.09	0.31	amihud_21 z=+0.8; log_dvol_63 z=-0.1; mom_12_1 z=+0.7	nan
HWM	Industrials	high	0.43	0.47	0.49	-1.3	-0.95	-3.95	0.55	1.63	2.16	2.99	mom_12_1 z=-1.3; amihud_21 z=-1.6; dn_vol_63 z=+2.4	nan
KIM	Real Estate	low	0.43	0.39	0.77	0.46	-0.21	-0.87	-1.16	0.41	-0.99	0.75	days_since_20pct_dd z=+2.3; dvol_trend_21_126 z=+0.4; nr7_count_20 z=+2.1	nan
FIX	Industrials	high	0.43	0.46	0.63	-0.31	-1.19	-0.79	-0.97	-1.37	0.14	0.36	days_since_52w_high z=+2.5; amihud_21 z=-1.5; days_since_2x_low z=+2.2	nan
WFC	Financial	mid	0.43	0.44	0.67	-0.75	-1.1	-1.22	-0.24	-0.61	0.5	1.08	corr_spy_63 z=-0.1; up_day_frac_63 z=+0.1; rs_sec_63 z=-0.6	nan
HON	Industrials	mid	0.42	0.43	0.72	-0.59	-1.34	-1.69	-1.09	-1.5	0.95	0.25	days_since_20pct_dd z=-1.7; clv_21 z=+1.3; log_dvol_63 z=-0.4	nan
MRSB	Financial	low	0.42	0.39	0.53	-0.5	0.37	-0.51	-0.99	-0.46	0.56	-0.67	days_since_20pct_dd z=-1.2; dvol_trend_21_126 z=+0.2; nr7_count_20 z=-0.8	nan
EMR	Industrials	high	0.42	0.46	0.87	-0.27	0.27	1.15	-0.94	-0.78	-0.4	-0.12	amihud_21 z=+0.1; up_day_frac_63 z=+1.4; log_dvol_63 z=+0.7	nan
LDOS	Technology	mid	0.42	0.42	0.24	-1.76	-2.63	-1.81	-0.39	0.06	-0.5	-0.25	corr_spy_63 z=+0.2; clv_21 z=-1.9; days_since_20pct_dd z=-0.9	nan
AMT	Real Estate	low	0.42	0.38	0.57	-0.32	0.36	-0.07	-1.05	-1.03	-1.42	-0.15	dvol_trend_21_126 z=-0.9; streak_now z=+0.8; days_since_20pct_dd z=+0.1	nan
MPC	Energy	low	0.41	0.37	0.43	2.45	0.82	1.06	-0.09	1.99	0.09	-1.76	days_since_20pct_dd z=+2.2; days_since_2x_low z=-1.3; dvol_trend_21_126 z=+3.3	nan
NKE	Consumer Cyclical	mid	0.41	0.42	0.06	-2.13	-0.21	-2.18	-0.83	2.62	-1.03	-0.02	log_dvol_63 z=+2.1; up_day_frac_63 z=-1.7; sharpe_21 z=-1.9	nan
CCI	Real Estate	low	0.41	0.37	0.2	-0.76	0.5	-1.32	-0.77	0.04	-0.6	0.63	dvol_trend_21_126 z=-0.8; nr7_count_20 z=-1.0; streak_now z=+0.8	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
EG	Financial	low	0.41	0.36	0.84	0.8	0.35	0.42	-0.85	0.61	-1.9	-0.31	dvol_trend_21_126 z=+0.9; streak_now z=-1.0; nr7_count_20 z=+0.9	nan
BAC	Financial	mid	0.41	0.41	0.59	0.25	-0.25	-1.36	0.33	0.7	0.54	1.93	base_depth_126 z=-1.9; rs_sec_63 z=-1.2; sharpe_21 z=-2.5	nan
CTAS	Industrials	low	0.4	0.36	0.51	-0.81	-0.55	-0.23	-0.31	-0.36	-0.37	-0.6	days_since_20pct_dd z=-1.8; dvol_trend_21_126 z=-0.1; streak_now z=+0.3	nan
RJF	Financial	mid	0.4	0.41	0.61	-0.4	-0.13	-0.43	-0.68	-0.2	-1.27	0.3	sharpe_21 z=-2.3; up_day_frac_63 z=-2.2; days_since_20pct_dd z=+1.9	nan
COST	Consumer Defensive	low	0.4	0.35	0.56	-1.01	-0.71	-1.54	-0.73	0.07	1.22	0.4	streak_now z=+1.2; nr7_count_20 z=+0.6; days_since_20pct_dd z=-0.7	nan
CSGP	Real Estate	mid	0.4	0.4	0.11	-2.02	-1.48	-1.91	-1.04	-0.66	-3.05	-0.98	up_day_frac_63 z=-2.3; log_dvol_63 z=-0.2; clv_21 z=-0.1	nan
GM	Consumer Cyclical	high	0.4	0.45	0.68	0.63	-0.96	-0.14	0.79	0.31	1.13	0.25	amihud_21 z=+1.2; log_dvol_63 z=-0.4; mom_12_1 z=+0.6	nan
SLB	Energy	mid	0.39	0.39	0.34	2.46	0.88	0.26	-0.49	-0.8	0.2	-1.55	sharpe_21 z=-1.5; clv_21 z=-1.5; days_since_20pct_dd z=+0.5	nan
SYF	Financial	high	0.39	0.45	0.45	-0.7	-1.23	-0.88	-0.76	-1.12	0.61	1	days_since_52w_high z=+1.8; log_dvol_63 z=+0.5; up_day_frac_63 z=+0.1	nan
CL	Consumer Defensive	low	0.39	0.34	0.78	0.61	0.37	-0.23	-1.25	-0.49	-0.5	0.51	nr7_count_20 z=+1.5; dvol_trend_21_126 z=+0.3; streak_now z=+0.4	nan
GPN	Technology	high	0.39	0.44	0.45	0.87	0.55	0.83	-0.63	-1.13	-0.15	-0.67	amihud_21 z=+1.0; mom_12_1 z=+0.9; days_since_52w_high z=-1.6	nan
ECL	Basic Materials	mid	0.39	0.39	0.62	-1.17	-1.01	-0.8	-0.87	-1.3	-0.54	0.01	days_since_20pct_dd z=-1.7; sharpe_21 z=-0.4; up_day_frac_63 z=-1.5	nan
BKR	Energy	mid	0.38	0.38	0.36	0.35	-0.46	-1.13	-0.21	-0.15	0.3	-0.02	days_since_20pct_dd z=-1.3; rs_sec_63 z=-1.1; base_depth_126 z=-1.2	nan
PNC	Financial	mid	0.38	0.38	0.59	0.34	-0.19	-0.85	-0.28	-0.31	1.6	2.42	rs_sec_63 z=-1.4; corr_spy_63 z=-0.2; days_since_20pct_dd z=+1.9	nan
KHC	Consumer Defensive	low	0.38	0.34	0.22	1.16	0.44	-0.64	-1.69	1.19	-0.19	0.07	dvol_trend_21_126 z=+2.4; streak_now z=-2.9; days_since_20pct_dd z=+0.6	nan
HBAN	Financial	mid	0.38	0.37	0.35	-0.61	-0.26	-1.05	-0.05	0.17	-0.28	1.7	rs_sec_63 z=-1.5; corr_spy_63 z=+0.4; base_depth_126 z=-1.3	nan
ANET	Technology	high	0.38	0.43	0.87	-0.85	-0.34	0.76	-1.21	-1.51	0.67	-0.53	days_since_ignition z=+1.5; days_since_2x_low z=+1.6; dn_vol_63 z=-0.3	nan
ALB	Basic Materials	high	0.37	0.43	0.19	0.55	-0.47	-0.78	-0.95	-0.32	-0.34	0.23	log_dvol_63 z=-0.2; amihud_21 z=-0.0; dn_vol_63 z=-1.5	nan
ADSK	Technology	mid	0.37	0.36	0.24	-1.93	0.16	-1.72	-0.15	0.05	-1.98	-0.55	days_since_20pct_dd z=-1.0; corr_spy_63 z=-1.8; sharpe_21 z=-0.5	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
FITB	Financial	mid	0.37	0.36	0.52	0.15	-0.5	-0.51	-0.65	-0.61	0.26	1.82	corr_spy_63 z=+0.0; rs_sec_63 z=-1.2; base_depth_126 z=-1.0	nan
GE	Industrials	high	0.37	0.42	0.58	-2.04	-0.65	-3.05	-1.01	-0.05	1.89	1.85	mom_12_1 z=-2.0; amihud_21 z=-1.5; days_since_2x_low z=+1.4	nan
SBUX	Consumer Cyclical	mid	0.37	0.35	0.41	1.58	0.09	0.24	-0.85	-0.54	0.77	0.29	log_dvol_63 z=-0.7; days_since_20pct_dd z=+2.4; sharpe_21 z=-2.4	nan
PEP	Consumer Defensive	low	0.36	0.33	0.38	0.34	-0.54	-1.94	-1.23	0.56	-0.9	0.89	days_since_20pct_dd z=-1.3; dvol_trend_21_126 z=+0.5; streak_now z=-0.4	nan
SBAC	Real Estate	low	0.36	0.33	0.13	0.67	-0.67	-1.29	-0.63	-0.2	-1.38	-0.12	dvol_trend_21_126 z=-0.6; streak_now z=+0.9; nr7_count_20 z=-0.9	nan
CLX	Consumer Defensive	mid	0.36	0.35	0.12	-0.77	-0.38	-1.56	-0.29	0.63	-1.2	0	corr_spy_63 z=+1.6; base_depth_126 z=-1.2; log_dvol_63 z=+1.3	nan
PLD	Real Estate	mid	0.36	0.34	0.35	0.83	-0.03	-0.82	-0.89	-0.27	1.12	0.78	days_since_20pct_dd z=+2.5; corr_spy_63 z=-1.7; up_day_frac_63 z=-1.7	nan
CDW	Technology	mid	0.36	0.33	0.22	0.17	0.36	0.5	-0.27	-1.24	-0.66	-0.48	clv_21 z=-2.0; up_day_frac_63 z=-1.8; base_depth_126 z=+0.0	nan
L	Financial	low	0.35	0.32	0.86	-2.01	-0.26	-3.29	-1.09	2.36	0.42	1.79	dvol_trend_21_126 z=+1.2; days_since_20pct_dd z=+0.8; streak_now z=-0.6	nan
GL	Financial	mid	0.35	0.33	0.71	0.62	0.19	-0.17	-0.35	-0.25	-0.68	1.1	corr_spy_63 z=-0.0; sharpe_21 z=-2.2; up_day_frac_63 z=-1.7	nan
EOG	Energy	low	0.35	0.31	0.66	2.45	-0.11	0.75	0.07	-1.01	0.79	0.03	streak_now z=+1.1; days_since_20pct_dd z=+0.2; nr7_count_20 z=-0.8	nan
FDX	Industrials	high	0.35	0.42	0.46	1.4	-0.25	-0.42	-0.38	-0.62	-0.73	0.13	mom_12_1 z=+1.4; amihud_21 z=+0.3; dn_vol_63 z=-0.5	nan
IBM	Technology	high	0.35	0.41	0.13	-2.14	0.01	-2.4	-0.63	-0.71	-1.01	2.07	mom_12_1 z=-2.1; amihud_21 z=-0.1; dn_vol_63 z=+2.7	nan
ARE	Real Estate	mid	0.34	0.32	0.19	-0.77	0.57	-0.91	-0.63	-1.05	-0.2	-0.56	log_dvol_63 z=-1.2; corr_spy_63 z=-2.0; rs_sec_63 z=+0.7	nan
APD	Basic Materials	low	0.34	0.31	0.83	0.76	-0.49	0.29	-0.62	-0.45	0.02	0.3	days_since_20pct_dd z=+0.9; nr7_count_20 z=+0.3; streak_now z=+0.8	nan
APH	Technology	high	0.34	0.4	0.69	-0.64	-1.09	0.75	-0.8	-1.96	-1.42	0.03	days_since_52w_high z=+1.7; up_day_frac_63 z=-1.3; days_since_ignition z=+1.1	nan
FERG	Industrials	high	0.34	0.4	0.16	-1.11	-1.26	-0.97	-1.22	-0.5	1.17	-0.13	mom_12_1 z=-1.1; days_since_52w_high z=+1.6; parkinson_cc_20 z=+2.1	nan
STZ	Consumer Defensive	low	0.34	0.3	0.28	0.3	-0.55	-0.83	-0.64	-0.3	-0.27	0.87	days_since_20pct_dd z=-0.8; dvol_trend_21_126 z=-0.4; nr7_count_20 z=-0.9	nan
COR	Healthcare	low	0.33	0.3	0.52	-0.93	-0.93	-1.28	-0.34	-0.47	-1.03	-0.09	days_since_20pct_dd z=-2.3; dvol_trend_21_126 z=-0.7; nr7_count_20 z=-1.5	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
ALL	Financial	low	0.33	0.29	0.31	0.18	-0.09	-3.56	0.13	3.32	-2.05	1.31	dvol_trend_21_126 z=+2.1; days_since_20pct_dd z=+1.0; streak_now z=+0.3	nan
OMC	Communication Services	mid	0.33	0.32	0.33	0.72	0.52	-0.07	-0.9	-1.03	-0.34	0.02	clv_21 z=-1.9; days_since_20pct_dd z=+1.4; log_dvol_63 z=+0.3	nan
ORCL	Technology	high	0.33	0.39	0.18	-2.37	-0.02	-1.63	-0.8	0.13	0.43	0.06	mom_12_1 z=-2.4; days_since_52w_high z=+2.5; amihud_21 z=-1.4	nan
JCI	Industrials	high	0.33	0.39	0.88	0.06	-0.46	1.05	-0.01	-0.96	-0.98	-0.27	amihud_21 z=-0.1; days_since_52w_high z=-0.8; log_dvol_63 z=+0.7	nan
DD	Basic Materials	high	0.32	0.38	0.39	0.51	-0.51	-0.9	-0.95	-1.02	0.05	0.35	log_dvol_63 z=-0.6; amihud_21 z=+0.1; mom_12_1 z=+0.5	nan
RL	Consumer Cyclical	high	0.32	0.37	0.52	-2.16	-1.18	-0.78	-0.64	0.45	1.42	1.04	mom_12_1 z=-2.2; amihud_21 z=-1.4; days_since_52w_high z=+1.4	nan
TPL	Energy	mid	0.32	0.31	0.15	-0.32	-1.33	-0.77	-0.31	-0.24	-0.13	0.66	rs_sec_63 z=-1.3; corr_spy_63 z=-0.5; log_dvol_63 z=+0.3	nan
NCLH	Consumer Cyclical	high	0.32	0.37	0.01	-1.8	-0.61	-1.47	-1.49	0.17	-0.32	0.59	mom_12_1 z=-1.8; log_dvol_63 z=-0.2; amihud_21 z=-1.4	nan
HLT	Consumer Cyclical	mid	0.32	0.3	0.49	-0.48	-1.5	-1.31	-0.92	-0.58	1.59	1.33	corr_spy_63 z=-2.3; base_depth_126 z=-1.3; log_dvol_63 z=+1.3	nan
GNRC	Industrials	high	0.31	0.36	0.35	-0.44	-0.86	-0.65	1.01	1.71	-0.02	-0.6	log_dvol_63 z=+1.6; amihud_21 z=-0.3; up_day_frac_63 z=-1.9	nan
WSM	Consumer Cyclical	high	0.31	0.36	0.79	-0.8	0.1	0.63	-0.69	0.31	0.41	0.13	log_dvol_63 z=+0.3; dn_vol_63 z=-0.9; days_since_2x_low z=+1.1	nan
ICE	Financial	mid	0.31	0.3	0.42	-0.66	-2.13	-0.61	-0.95	-1.72	-0.88	-0.82	rs_sec_63 z=+1.8; corr_spy_63 z=-2.1; days_since_20pct_dd z=-1.1	nan
UPS	Industrials	mid	0.31	0.29	0.3	2.37	0.4	0.14	-0.34	-1.02	0.2	0.56	log_dvol_63 z=-0.7; sharpe_21 z=-1.4; corr_spy_63 z=-1.0	nan
KEY	Financial	mid	0.31	0.28	0.45	0.08	-0.52	-0.34	0.07	0.37	0.19	1.7	corr_spy_63 z=+0.2; rs_sec_63 z=-1.3; days_since_20pct_dd z=+2.2	nan
VRT	Industrials	high	0.3	0.35	0.55	-0.72	-0.98	-1.31	-0.33	-0.82	2.52	1.48	days_since_52w_high z=+1.4; log_dvol_63 z=+0.7; days_since_2x_low z=-1.2	nan
ECHO	Communication Services	high	0.3	0.34	0.25	-0.65	-0.34	-0.6	-0.62	-0.93	-0.49	-0.14	dn_vol_63 z=-1.3; days_since_ignition z=+1.3; days_since_2x_low z=+1.3	nan
COF	Financial	high	0.3	0.34	0.29	-1.11	0.27	-1.55	-0.98	-0.91	0.06	0.48	mom_12_1 z=-1.1; days_since_52w_high z=+1.9; log_dvol_63 z=+0.7	nan
EIX	Utilities	low	0.3	0.28	0.01	0.05	0.16	-1.2	0.05	0.85	2.41	2.19	streak_now z=+2.0; dvol_trend_21_126 z=+0.4; days_since_20pct_dd z=-1.0	nan
TMUS	Communication Services	low	0.3	0.28	0.24	-1.18	-1.36	-1.41	-0.1	-0.33	0.69	1.11	days_since_20pct_dd z=-1.5; dvol_trend_21_126 z=-0.8; nr7_count_20 z=-0.9	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
PHM	Consumer Cyclical	high	0.29	0.33	0.25	-1.04	-0.83	-0.73	-1.16	0.16	1.13	1.23	log_dvol_63 z=-0.7; amihud_21 z=+0.1; mom_12_1 z=-1.0	nan
ADM	Consumer Defensive	low	0.29	0.27	0.69	1.61	0.41	0.98	-0.33	-0.04	-1.08	-0.62	days_since_20pct_dd z=+2.5; streak_now z=+1.2; nr7_count_20 z=+1.1	nan
CASY	Consumer Cyclical	low	0.29	0.27	0.02	-0.41	-0.48	-4.79	1.68	2.04	0.62	2.54	days_since_20pct_dd z=-6.0; days_since_2x_low z=-1.7; streak_now z=-0.9	nan
LYV	Communication Services	mid	0.29	0.28	0.53	-0.66	-0.56	-0.01	-1.61	-1.2	0.85	0.81	corr_spy_63 z=-1.9; rs_sec_63 z=-0.9; log_dvol_63 z=+0.5	nan
TYL	Technology	low	0.29	0.26	0.1	-0.96	0.16	-1.1	-0.33	-0.34	-2	-0.05	days_since_20pct_dd z=-1.2; dvol_trend_21_126 z=-0.9; streak_now z=+0.5	nan
BSX	Healthcare	low	0.28	0.25	0.06	-1.99	-0.68	-2.32	-0.33	0.45	-0.67	0.36	days_since_20pct_dd z=-1.8; streak_now z=-2.1; dvol_trend_21_126 z=-0.7	nan
RF	Financial	mid	0.28	0.27	0.47	0.19	-0.35	-0.51	-0.53	-0.85	-0.89	1	days_since_20pct_dd z=+2.1; sharpe_21 z=-2.3; clv_21 z=-2.3	nan
MLM	Basic Materials	mid	0.28	0.27	0.18	-1.73	-1.71	-2.57	-1.13	2.61	0.78	1.57	rs_sec_63 z=-1.6; log_dvol_63 z=+2.0; base_depth_126 z=-1.8	nan
CHRW	Industrials	mid	0.28	0.26	0.23	-0.44	0.06	-2.13	0.4	-0.37	0.31	1.47	base_depth_126 z=-2.7; rs_sec_63 z=-1.4; clv_21 z=+1.3	nan
PNR	Industrials	high	0.28	0.33	0.01	-2.5	-1.39	-3.1	-0.86	0.29	-0.46	1.83	mom_12_1 z=-2.5; days_since_52w_high z=+2.5; dn_vol_63 z=+2.2	nan
TFC	Financial	mid	0.27	0.25	0.46	0.11	-0.83	-0.41	-0.33	0.1	0.95	1.47	clv_21 z=-2.4; sharpe_21 z=-2.1; rs_sec_63 z=-0.8	nan
GEHC	Healthcare	mid	0.27	0.25	0.2	-0.72	-1.6	-1.23	-0.8	-0.52	-1.84	-0.97	corr_spy_63 z=-2.4; log_dvol_63 z=-0.2; days_since_20pct_dd z=-0.8	nan
UAL	Industrials	high	0.27	0.32	0.36	-0.46	-0.7	-0.21	-0.92	-0.99	1.68	0.59	log_dvol_63 z=-0.0; amihud_21 z=-0.1; days_since_52w_high z=-0.4	nan
LLY	Healthcare	mid	0.27	0.24	0.54	0.03	0.47	-0.14	-1.4	-1.25	1	0.34	rs_sec_63 z=-1.2; corr_spy_63 z=-2.0; sharpe_21 z=-0.6	nan
ARES	Financial	high	0.27	0.31	0.2	-0.83	0.45	-1.35	-1.14	-1	-0.56	-0.84	log_dvol_63 z=+0.5; days_since_52w_high z=+1.5; dn_vol_63 z=-0.3	nan
PSA	Real Estate	mid	0.26	0.24	0.31	0.27	0.01	-0.58	-0.99	-1	0.76	1.17	rs_sec_63 z=-1.1; up_day_frac_63 z=-2.0; days_since_20pct_dd z=+1.5	nan
IDXX	Healthcare	high	0.26	0.31	0.27	-1.15	-1.8	-1.57	-0.74	-0.73	2.55	0.64	amihud_21 z=+0.2; mom_12_1 z=-1.1; log_dvol_63 z=+0.4	nan
APP	Communication Services	high	0.26	0.3	0.05	-1.56	-0.18	-2.34	-0.8	-0.23	-0.6	2.02	mom_12_1 z=-1.6; days_since_52w_high z=+2.7; up_day_frac_63 z=-2.7	nan
GRMN	Technology	high	0.26	0.3	0.6	-1.15	0.07	-0.09	-0.65	-0.62	-2.4	-1.01	mom_12_1 z=-1.2; up_day_frac_63 z=-2.9; dn_vol_63 z=-0.3	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
DHI	Consumer Cyclical	high	0.26	0.29	0.11	-1.15	-0.85	-0.54	-1.24	0.64	1.02	0.71	log_dvol_63 z=-0.6; amihud_21 z=-0.2; mom_12_1 z=-1.1	nan
AP0	Financial	high	0.25	0.28	0.17	-0.61	0.09	-0.78	-1.13	-0.88	-0.05	0.04	log_dvol_63 z=+0.2; amihud_21 z=-0.3; days_since_2x_low z=+1.3	nan
C00	Healthcare	mid	0.25	0.23	0.09	0.31	-0.2	-1.53	2.51	3.11	0.28	1.69	rs_sec_63 z=-1.8; base_depth_126 z=-1.3; clv_21 z=-0.2	nan
NVDA	Technology	high	0.25	0.28	0.86	-1.05	-0.97	1.29	-1.11	-1.01	-0.5	-0.39	log_dvol_63 z=-0.8; days_since_52w_high z=-1.1; dn_vol_63 z=-0.6	nan
KMB	Consumer Defensive	low	0.25	0.25	0.34	-0.45	0.67	-0.45	-0.53	-0.56	-0.05	0.39	days_since_20pct_dd z=-1.4; dvol_trend_21_126 z=+0.1; streak_now z=+0.8	nan
ISRG	Healthcare	high	0.25	0.27	0.22	-1.32	-1.32	-1.59	-0.18	-0.13	0.58	1.21	amihud_21 z=+0.5; mom_12_1 z=-1.3; dn_vol_63 z=+2.3	nan
TKO	Communication Services	mid	0.24	0.22	0.17	-1.25	-0.61	-0.97	-0.56	-0.1	-0.56	0.12	days_since_20pct_dd z=-1.2; rs_sec_63 z=-0.9; up_day_frac_63 z=-1.7	nan
STE	Healthcare	mid	0.24	0.22	0.27	-1.11	-0.27	-1.62	-1.06	-0.83	-2.38	-0.38	days_since_20pct_dd z=-1.6; up_day_frac_63 z=-2.2; corr_spy_63 z=-1.5	nan
AVGO	Technology	high	0.24	0.27	0.47	-2.62	-0.73	-1.74	-0.9	-0.2	-0.12	0.69	mom_12_1 z=-2.6; days_since_52w_high z=+2.0; amihud_21 z=-1.3	nan
LII	Industrials	high	0.24	0.26	0.04	-1.54	-0.41	-2.19	-1.42	-0.15	-0.08	3.45	mom_12_1 z=-1.5; dn_vol_63 z=+3.0; log_dvol_63 z=+0.3	nan
CMS	Utilities	low	0.24	0.24	0.39	-1.08	-0.7	-3.11	-1.04	1.26	0.47	3.13	days_since_20pct_dd z=-1.9; streak_now z=+1.1; dvol_trend_21_126 z=-0.1	nan
PAYX	Technology	low	0.23	0.23	0.04	0.15	0.7	-0.64	0.69	-0.09	0.1	0.52	dvol_trend_21_126 z=+0.8; days_since_20pct_dd z=-1.1; nr7_count_20 z=+0.9	nan
OTIS	Industrials	mid	0.23	0.21	0.16	-1.66	-0.9	-2.17	0.12	1.81	-0.05	-0.2	days_since_20pct_dd z=-1.9; base_depth_126 z=-1.4; corr_spy_63 z=-1.6	nan
TEL	Technology	high	0.23	0.25	0.36	-1.29	-1.34	-0.75	-0.74	-1.4	-2.7	0	mom_12_1 z=-1.3; amihud_21 z=-0.1; days_since_52w_high z=+1.2	nan
VMRK	Real Estate	low	0.23	0.23	0.39	0.06	0.18	-0.94	0.18	3.42	0.83	1.53	dvol_trend_21_126 z=+2.8; days_since_20pct_dd z=+0.2; streak_now z=+0.3	nan
HIG	Financial	low	0.23	0.22	0.62	-0.97	-0.92	-2.61	-0.02	1.61	0.69	0.52	dvol_trend_21_126 z=+0.3; streak_now z=+1.1; days_since_20pct_dd z=+0.0	nan
VST	Utilities	high	0.22	0.25	0.27	-1.35	-1.3	-1.25	-1.08	0.31	1.53	1.2	mom_12_1 z=-1.3; log_dvol_63 z=+0.3; days_since_52w_high z=+2.7	nan
ZTS	Healthcare	mid	0.22	0.21	0.12	-1.83	-0.16	-1.99	-0.66	-0.94	-0.46	-0.36	base_depth_126 z=-2.1; log_dvol_63 z=-0.4; corr_spy_63 z=-1.9	nan
MDT	Healthcare	low	0.22	0.22	0.37	-0.63	-0.25	-0.78	-0.32	4.61	0.34	-0.07	dvol_trend_21_126 z=+4.4; days_since_20pct_dd z=-1.0; nr7_count_20 z=+1.0	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
FSLR	Technology	high	0.22	0.24	0.22	-0.76	0.15	-1.13	0.31	0.23	1.12	0.89	amihud_21 z=+0.7; log_dvol_63 z=-0.3; days_since_52w_high z=-0.4	nan
BX	Financial	high	0.22	0.23	0.1	-1.24	0.52	-1.18	-0.9	-0.57	0.88	0.11	amihud_21 z=+0.2; mom_12_1 z=-1.2; log_dvol_63 z=+0.3	nan
HUBB	Industrials	high	0.21	0.23	0.41	-0.65	-1.83	-0.55	-0.92	-0.61	-0.03	0.07	log_dvol_63 z=+0.3; amihud_21 z=-0.4; dn_vol_63 z=-0.1	nan
KMI	Energy	low	0.21	0.21	0.79	-0.45	-1.02	-0.54	-0.44	0.53	0.43	0.17	dvol_trend_21_126 z=+0.8; days_since_20pct_dd z=+0.5; streak_now z=+1.0	nan
CBRE	Real Estate	high	0.21	0.22	0.08	-1.27	-1.12	-2.13	-0.36	0.25	1.06	0.11	mom_12_1 z=-1.3; days_since_52w_high z=+1.6; up_day_frac_63 z=-1.4	nan
AMZN	Consumer Cyclical	high	0.21	0.22	0.38	-0.58	-0.89	-0.62	-0.83	-1.05	-2.72	-1.75	log_dvol_63 z=+0.4; up_day_frac_63 z=-2.9; dn_vol_63 z=-0.4	nan
TXT	Industrials	mid	0.21	0.2	0.19	-1	-0.8	-1.88	-0.91	1.04	0.29	1.15	base_depth_126 z=-2.0; days_since_20pct_dd z=-1.3; rs_sec_63 z=-1.1	nan
HII	Industrials	mid	0.2	0.19	0.1	-0.43	-1.49	-1.93	-0.68	-0.11	-1.35	-0.61	corr_spy_63 z=+0.2; base_depth_126 z=-1.6; up_day_frac_63 z=-2.2	nan
UDR	Real Estate	low	0.2	0.2	0.45	0.16	0.45	-0.98	-0.86	0.34	-0.8	1.75	nr7_count_20 z=+0.8; dvol_trend_21_126 z=-0.1; streak_now z=+0.4	nan
KO	Consumer Defensive	low	0.2	0.2	0.96	2.46	0.79	-0.12	-1.29	-0.54	1.67	-0.34	nr7_count_20 z=+1.6; dvol_trend_21_126 z=-0.0; streak_now z=+0.3	nan
AXP	Financial	high	0.2	0.21	0.14	-1.32	-0.53	-1.87	-0.76	0.62	-0.48	1.24	mom_12_1 z=-1.3; amihud_21 z=-1.7; days_since_52w_high z=+1.9	nan
ABT	Healthcare	low	0.2	0.19	0.25	-1.21	0.9	-1.36	-0.44	-0.89	1.25	-1.22	dvol_trend_21_126 z=-1.2; days_since_20pct_dd z=-1.2; nr7_count_20 z=+0.8	nan
BA	Industrials	high	0.19	0.2	0.1	-0.39	-0.32	-0.38	-0.06	0.51	1.67	0.43	log_dvol_63 z=-0.8; amihud_21 z=+0.1; ignition_mult_10 z=-0.7	nan
CRH	Basic Materials	high	0.19	0.2	0.05	-2.57	-1.94	-3.22	-1.12	-0.25	1.3	2.17	mom_12_1 z=-2.6; days_since_52w_high z=+3.0; log_dvol_63 z=+0.4	nan
APTV	Consumer Cyclical	high	0.19	0.19	0.11	-1.6	0.53	-1.93	-0.63	0.52	-0.3	1.44	amihud_21 z=+0.9; log_dvol_63 z=-0.9; mom_12_1 z=-1.6	nan
RTX	Industrials	mid	0.19	0.19	0.36	-0.22	0.05	-1.61	-1.29	-0.75	0.61	0.33	base_depth_126 z=-1.8; sharpe_21 z=-3.0; corr_spy_63 z=-1.6	nan
XYL	Industrials	mid	0.19	0.18	0.18	-1.97	-1.68	-2.38	-0.08	1.86	0.17	0.75	corr_spy_63 z=-2.2; days_since_20pct_dd z=-1.1; base_depth_126 z=-1.4	nan
INVH	Real Estate	low	0.18	0.19	0.75	0.33	0.3	-0.48	-1.57	-0.34	-1.24	0.84	dvol_trend_21_126 z=+0.1; days_since_20pct_dd z=-0.8; streak_now z=+0.5	nan
VRTX	Healthcare	mid	0.18	0.18	0.55	1.26	0.85	-0.05	-0.88	-1.24	1	-0.1	sharpe_21 z=-2.3; rs_sec_63 z=-0.8; corr_spy_63 z=-1.6	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
XYZ	Technology	high	0.18	0.19	0.42	0.23	-0.2	0.9	-0.99	-0.93	-1.73	-0.08	log_dvol_63 z=-2.1; amihud_21 z=+0.4; days_since_52w_high z=-1.2	nan
BG	Consumer Defensive	low	0.18	0.18	0.13	1.32	-0.6	-0.45	-0.21	1.15	0.67	0.41	dvol_trend_21_126 z=+1.0; nr7_count_20 z=-1.0; streak_now z=+0.9	nan
SMCI	Technology	high	0.18	0.18	0.55	-0.62	-0.22	0.91	-0.65	-0.32	0.08	-1.47	log_dvol_63 z=-0.5; amihud_21 z=+0.3; dn_vol_63 z=-1.1	nan
NI	Utilities	low	0.17	0.17	0.36	-1.32	-0.95	-3.45	-0.97	1.02	0.07	2.45	days_since_20pct_dd z=-5.0; nr7_count_20 z=-0.9; streak_now z=+0.3	nan
BKNG	Consumer Cyclical	high	0.17	0.17	0.12	-1.38	0.22	-2.15	-0.37	0.61	-1.92	0.2	mom_12_1 z=-1.4; days_since_52w_high z=+2.1; log_dvol_63 z=+0.4	nan
VRSN	Technology	low	0.17	0.17	0.48	-0.1	0.42	0.86	-0.65	-0.69	1.65	-0.63	streak_now z=+1.5; nr7_count_20 z=+0.1; dvol_trend_21_126 z=-0.4	nan
INTU	Technology	mid	0.17	0.17	0.07	-1.75	-1.82	-1.91	-0.52	-0.43	-0.84	-0.67	days_since_20pct_dd z=-1.0; corr_spy_63 z=-1.7; clv_21 z=-1.6	nan
MAA	Real Estate	low	0.16	0.16	0.4	-0.1	0.38	-0.61	-0.94	1.08	-0.57	1.99	dvol_trend_21_126 z=+0.7; days_since_20pct_dd z=-0.1; streak_now z=+0.5	nan
GEN	Technology	mid	0.16	0.16	0.21	0.11	0.05	-1.65	1.44	2.75	0.5	1.52	rs_sec_63 z=-1.8; base_depth_126 z=-2.1; log_dvol_63 z=+1.7	nan
EL	Consumer Defensive	high	0.16	0.17	0.4	0.89	0.66	0.76	-0.67	-1.13	-0.35	-1.71	log_dvol_63 z=-2.1; amihud_21 z=+0.7; mom_12_1 z=+0.9	nan
AXON	Industrials	high	0.16	0.16	0.07	-1.29	-0.12	-1.58	-0.72	0.09	0.21	1.05	mom_12_1 z=-1.3; days_since_52w_high z=+2.3; dn_vol_63 z=+1.8	nan
VICI	Real Estate	low	0.16	0.16	0.36	-1.89	-0.37	-3.13	-0.65	-0.21	1.29	1.19	days_since_20pct_dd z=-6.0; streak_now z=-1.8; dvol_trend_21_126 z=+0.1	nan
PYPL	Financial	high	0.15	0.16	0.31	-0.41	0.32	-0.38	-0.67	-1.07	0.12	-0.96	amihud_21 z=+1.6; log_dvol_63 z=-1.5; ignition_mult_10 z=-0.5	nan
BNY	Financial	mid	0.15	0.16	0.63	0.83	-0.47	-2.28	-0.63	1.2	-0.39	0.86	base_depth_126 z=-3.0; rs_sec_63 z=-1.8; sharpe_21 z=-3.5	nan
ETN	Industrials	high	0.15	0.15	0.7	-0.93	0.01	-0.05	-0.08	-0.9	0.07	-0.06	amihud_21 z=+0.4; log_dvol_63 z=+0.4; mom_12_1 z=-0.9	nan
RSG	Industrials	low	0.15	0.15	0.85	-0.88	-1.25	-0.15	-0.73	-0.17	-1.07	0.17	days_since_20pct_dd z=-1.3; dvol_trend_21_126 z=-0.4; streak_now z=+0.6	nan
CVNA	Consumer Cyclical	high	0.15	0.14	0.21	-1.07	-1.62	-1.27	-0.44	-0.64	1.32	0.74	log_dvol_63 z=-0.3; amihud_21 z=+0.1; mom_12_1 z=-1.1	nan
TDG	Industrials	mid	0.14	0.15	0.26	-1.41	-0.84	-2.31	-1	0.13	0.36	0.8	days_since_20pct_dd z=-2.0; rs_sec_63 z=-1.3; base_depth_126 z=-1.9	nan
SPG	Real Estate	mid	0.14	0.15	0.51	-0.44	-0.93	-1.77	-0.97	-1.07	1.28	1.39	base_depth_126 z=-1.8; rs_sec_63 z=-0.9; up_day_frac_63 z=-2.1	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
MSFT	Technology	high	0.14	0.14	0.52	-0.85	0.47	0.71	-0.53	-1.45	-0.94	-2.31	amihud_21 z=+0.1; days_since_52w_high z=+1.9; up_day_frac_63 z=+0.6	nan
PEG	Utilities	low	0.14	0.14	0.34	-1.18	-0.98	-2.32	-0.42	1.83	0.01	1.57	days_since_20pct_dd z=-3.2; dvol_trend_21_126 z=+0.6; streak_now z=+1.1	nan
ALGN	Healthcare	high	0.14	0.13	0.04	1.43	-0.72	0.25	-0.5	-0.63	-0.8	0.45	log_dvol_63 z=-2.3; amihud_21 z=+0.6; mom_12_1 z=+1.4	nan
SNA	Industrials	mid	0.13	0.14	0.48	-0.35	-0.09	-1.28	-1.44	-0.79	1.17	0.23	clv_21 z=-2.5; base_depth_126 z=-1.5; up_day_frac_63 z=-2.1	nan
CMG	Consumer Cyclical	mid	0.13	0.13	0.16	-0.38	-0.24	-0.32	-0.62	-0.73	1.17	-0.01	corr_spy_63 z=-2.3; log_dvol_63 z=-0.6; sharpe_21 z=-2.0	nan
HOOD	Financial	high	0.13	0.13	0.32	-1.18	-0.13	-0.36	-1.28	-0.79	-0.63	-0.59	days_since_52w_high z=+2.9; mom_12_1 z=-1.2; up_day_frac_63 z=-1.7	nan
OXY	Energy	low	0.13	0.14	0.47	2.03	-1.04	0.71	-0.15	-0.68	0.59	-0.53	days_since_2x_low z=+0.3; streak_now z=+1.8; nr7_count_20 z=-1.5	nan
FISV	Technology	mid	0.13	0.13	0.08	-1.22	-1.02	nan	nan	nan	nan	-0.17	log_dvol_63 z=-1.0; days_since_20pct_dd z=-1.0; clv_21 z=-1.7	nan
XOM	Energy	low	0.12	0.13	0.7	2.1	-0.63	0.6	-0.48	-1.18	-0.42	-1.33	days_since_20pct_dd z=-2.7; days_since_2x_low z=+0.2; nr7_count_20 z=-1.6	nan
COP	Energy	low	0.12	0.13	0.49	2.39	0.04	0.69	-0.07	-0.84	-0.13	-0.48	days_since_2x_low z=+0.2; days_since_20pct_dd z=-0.0; dvol_trend_21_126 z=-0.1	nan
DOC	Real Estate	mid	0.12	0.12	0.28	0.8	0.12	-0.26	-0.69	-1.4	-0.32	1.15	clv_21 z=-2.1; sharpe_21 z=-2.1; corr_spy_63 z=-1.7	nan
RDDT	Communication Services	high	0.12	0.12	0.09	-1.14	-0.48	-0.43	-1.34	-0.22	-1.71	0.97	log_dvol_63 z=+0.2; mom_12_1 z=-1.1; amihud_21 z=-0.4	nan
SYF	Consumer Defensive	low	0.12	0.12	0.6	-0.42	0	-1.08	0.06	1.21	-0.63	0.84	dvol_trend_21_126 z=+1.4; nr7_count_20 z=+1.5; streak_now z=-0.5	nan
LULU	Consumer Cyclical	high	0.11	0.11	0.03	-0.48	-0.75	-1.09	-0.82	0.82	-1.03	0.67	log_dvol_63 z=-1.8; amihud_21 z=+0.6; dn_vol_63 z=+0.9	nan
GD	Industrials	mid	0.11	0.12	0.38	-0.52	0.2	-2.03	-0.95	0.15	0.84	1.62	base_depth_126 z=-2.1; sharpe_21 z=-2.8; clv_21 z=-2.0	nan
TTWO	Communication Services	mid	0.11	0.11	0.15	-1.89	0.2	-2.18	-0.43	0.36	-0.89	0.56	rs_sec_63 z=-1.8; base_depth_126 z=-2.2; days_since_20pct_dd z=-1.5	nan
BLK	Financial	high	0.11	0.11	0.3	-1.39	0.21	-0.48	-0.67	-1.29	-2.3	-0.97	amihud_21 z=+0.7; mom_12_1 z=-1.4; days_since_52w_high z=+2.1	nan
TSN	Consumer Defensive	low	0.11	0.11	0.27	-0.4	-0.16	-1.72	-1.54	0.91	0.94	0.91	days_since_20pct_dd z=-1.6; dvol_trend_21_126 z=+0.2; streak_now z=-0.4	nan
HAL	Energy	mid	0.1	0.1	0.13	1.27	0.14	-0.19	-0.76	-1.03	-0.35	-0.14	rs_sec_63 z=-1.2; up_day_frac_63 z=-1.7; sharpe_21 z=-1.9	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
KKR	Financial	high	0.1	0.1	0.13	-0.95	0.23	-1.4	-0.97	-0.67	0.16	0.14	amihud_21 z=+0.4; log_dvol_63 z=-0.1; mom_12_1 z=-0.9	nan
NRG	Utilities	high	0.1	0.1	0.06	-2.16	-1.36	-3.43	-1.26	1.12	-1.1	1.49	mom_12_1 z=-2.2; amihud_21 z=+0.1; days_since_52w_high z=+2.2	nan
DASH	Consumer Cyclical	high	0.1	0.09	0.33	-1.57	-0.02	-1.2	-0.36	-0.42	0.43	0.48	mom_12_1 z=-1.6; days_since_52w_high z=+2.9; amihud_21 z=-0.1	nan
ROL	Consumer Cyclical	low	0.1	0.11	0.03	-3.04	-0.18	-3.59	0.68	2.35	-0.3	1.6	days_since_20pct_dd z=-1.4; dvol_trend_21_126 z=+0.6; streak_now z=-0.9	nan
IP	Consumer Cyclical	high	0.09	0.08	0.02	-1.02	-0.15	-1.53	-0.93	-0.38	0.36	0.02	amihud_21 z=+0.4; log_dvol_63 z=+0.1; days_since_ignition z=-1.3	nan
EME	Industrials	high	0.09	0.08	0.34	-1.44	-1.44	-0.98	-0.78	-0.38	-1.23	-0.72	mom_12_1 z=-1.4; days_since_52w_high z=+2.4; days_since_2x_low z=+2.9	nan
FDS	Financial	low	0.09	0.1	0.16	1.02	0.76	0.41	-0.29	-0.55	-1.01	-0.63	days_since_20pct_dd z=-1.1; dvol_trend_21_126 z=+0.1; nr7_count_20 z=+0.0	nan
LEN	Consumer Cyclical	high	0.09	0.07	0.08	-1.18	-1.28	-1.41	1.02	1.53	0.27	0.64	log_dvol_63 z=-1.3; mom_12_1 z=-1.2; amihud_21 z=-0.2	nan
VMC	Basic Materials	mid	0.09	0.1	0.25	-1.95	-0.74	-2.59	-1.16	0.35	-0.2	1.03	base_depth_126 z=-2.4; rs_sec_63 z=-1.6; days_since_20pct_dd z=-1.3	nan
MCD	Consumer Cyclical	low	0.08	0.1	0.18	-1.63	-0.63	-3.46	0.26	2.74	-0.56	1.36	days_since_20pct_dd z=-3.1; dvol_trend_21_126 z=+1.2; streak_now z=+1.3	nan
HD	Consumer Cyclical	mid	0.08	0.09	0.14	-1.71	-0.76	-2.18	-1.06	1.26	-0.77	1.48	rs_sec_63 z=-1.4; base_depth_126 z=-1.7; clv_21 z=-2.8	nan
BXP	Real Estate	mid	0.08	0.08	0.28	-0.77	-0.49	-0.5	-0.55	-0.74	0.76	0.32	corr_spy_63 z=-2.0; sharpe_21 z=-2.3; clv_21 z=-2.3	nan
CEG	Utilities	high	0.08	0.07	0.28	-1.56	-1.02	-1.44	-0.76	-0.4	-0.19	0.18	mom_12_1 z=-1.6; days_since_52w_high z=+2.9; log_dvol_63 z=+0.4	nan
DECK	Consumer Cyclical	high	0.08	0.06	0.02	-0.81	-0.79	-0.43	-0.98	0.69	-1.47	0.82	log_dvol_63 z=-0.7; amihud_21 z=-0.5; up_day_frac_63 z=-1.6	nan
MCO	Financial	mid	0.07	0.08	0.38	-0.69	-0.11	-1.61	-0.83	-0.81	0.22	0.41	sharpe_21 z=-2.3; rs_sec_63 z=-0.9; base_depth_126 z=-1.1	nan
PLTR	Technology	high	0.07	0.05	0.51	-1.26	-0.09	0.58	-1.49	-1.34	-1.05	-1.48	days_since_52w_high z=+3.0; mom_12_1 z=-1.3; log_dvol_63 z=+0.3	nan
UBER	Technology	high	0.07	0.05	0.07	-1.3	-0.68	-1.69	-1.2	0.16	-0.72	0.42	log_dvol_63 z=-0.4; mom_12_1 z=-1.3; days_since_52w_high z=+2.5	nan
TT	Industrials	high	0.07	0.04	0.51	-1.12	-1.44	0.03	-0.75	-0.64	-0.95	0.53	log_dvol_63 z=+0.2; mom_12_1 z=-1.1; ignition_mult_10 z=+1.1	nan
PCG	Utilities	low	0.07	0.09	0.01	-0.86	-0.56	-1.92	0.12	1.48	2.09	2.44	dvol_trend_21_126 z=+0.7; days_since_20pct_dd z=-1.0; streak_now z=+0.7	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
HCA	Healthcare	low	0.06	0.08	0.08	-1.16	-2.3	-0.93	-0.39	0.46	-2.41	-0.46	days_since_20pct_dd z=-1.0; nr7_count_20 z=+0.9; dvol_trend_21_126 z=+0.0	nan
LOW	Consumer Cyclical	mid	0.06	0.07	0.07	-1.73	-1.15	-2.95	-1.79	0.95	-1.2	0.98	base_depth_126 z=-2.2; rs_sec_63 z=-1.1; clv_21 z=-2.2	nan
ERIE	Financial	low	0.06	0.08	0.19	-0.67	-0.65	-0.55	-0.78	-0.42	0.1	0.41	days_since_2x_low z=+0.9; nr7_count_20 z=+1.8; dvol_trend_21_126 z=-0.5	nan
SNPS	Technology	high	0.06	0.04	0.37	-0.75	-0.19	0.9	0.05	0.02	-1.21	-1.43	log_dvol_63 z=+0.3; amihud_21 z=-0.2; days_since_ignition z=-2.1	nan
J	Industrials	high	0.06	0.03	0.42	-1.32	0.31	-0.69	-0.64	-1.2	-0.62	-0.55	amihud_21 z=+0.6; mom_12_1 z=-1.3; days_since_52w_high z=+2.6	nan
CHTR	Communication Services	mid	0.05	0.07	0.01	-1.05	0.03	-1.55	-0.41	0.21	-0	-0.23	log_dvol_63 z=-1.3; base_depth_126 z=-2.0; sharpe_21 z=-2.4	nan
NFLX	Communication Services	mid	0.05	0.06	0.08	-1.73	-1.2	-2.17	-0.51	-0.37	-0.43	0.72	base_depth_126 z=-2.2; days_since_20pct_dd z=-1.0; log_dvol_63 z=-0.3	nan
EXE	Energy	low	0.05	0.07	0.3	-0.88	-1.25	-2.05	-0.36	-0.29	0.08	-0.15	days_since_2x_low z=+0.2; days_since_20pct_dd z=-1.1; nr7_count_20 z=-0.8	nan
KR	Consumer Defensive	low	0.05	0.07	0.1	-1.52	-0.93	-2.04	0.13	0.76	0.7	0.31	days_since_20pct_dd z=-2.1; dvol_trend_21_126 z=+0.4; streak_now z=-0.9	nan
SPGI	Financial	mid	0.05	0.05	0.21	-0.56	-0.15	-1.64	-0.8	-0.68	-0.98	0.18	days_since_20pct_dd z=-1.6; sharpe_21 z=-2.5; clv_21 z=-2.4	nan
WY	Real Estate	mid	0.04	0.05	0.03	-0.09	-0.35	-2.24	-0.22	0.91	0.14	0.43	base_depth_126 z=-2.8; sharpe_21 z=-2.7; rs_sec_63 z=-1.2	nan
TJX	Consumer Cyclical	low	0.04	0.06	0.14	-3.63	-0.94	-3.49	0.49	3.33	-0.27	1.54	days_since_20pct_dd z=-4.0; dvol_trend_21_126 z=+1.3; nr7_count_20 z=-1.0	nan
LHX	Industrials	mid	0.04	0.04	0.04	-1.25	-1.62	-2.73	-1.58	0.49	0.48	1.96	base_depth_126 z=-2.8; rs_sec_63 z=-1.3; days_since_20pct_dd z=-1.1	nan
TSLA	Consumer Cyclical	high	0.04	0.02	0.17	-1.14	-0.49	0.06	-0.92	-1.16	-0.99	1.07	log_dvol_63 z=-2.3; amihud_21 z=+0.4; mom_12_1 z=-1.1	nan
FICO	Technology	mid	0.04	0.04	0.05	-1.4	-0.21	-2.39	2.84	5.11	0.07	2.22	rs_sec_63 z=-2.4; base_depth_126 z=-2.4; corr_spy_63 z=-1.9	nan
RCL	Consumer Cyclical	high	0.03	0.02	0.24	-1.95	-0.86	-0.94	-0.32	2.06	-1.71	0.18	mom_12_1 z=-2.0; days_since_52w_high z=+2.5; amihud_21 z=-1.4	nan
EFX	Industrials	mid	0.03	0.03	0.04	-0.79	-0.59	-1.9	-0.96	0.36	-0.4	0.12	sharpe_21 z=-2.6; base_depth_126 z=-1.5; clv_21 z=-2.0	nan
UHS	Healthcare	low	0.03	0.05	0.2	-1.22	-1.87	-0.94	-0.62	-0.24	-0.17	-0.36	days_since_2x_low z=+0.6; streak_now z=+1.2; days_since_20pct_dd z=-0.7	nan
WYNN	Consumer Cyclical	mid	0.03	0.02	0.02	-1.64	-1.08	-2.73	-0.74	2.15	-0.35	1.11	log_dvol_63 z=-1.0; base_depth_126 z=-1.8; rs_sec_63 z=-1.0	nan

ticker	Sector	beta_bucket	state_score_pooled	state_score	level_score	z_mom_12_1	z_rs_lead_126	z_dist_52w_high	z_rv20_pct_252	z_vol_dry_20_250	z_obv_price_div_63	z_dn_up_vol_asym_63	top_states	note
COIN	Financial	high	0.03	0.01	0.15	-1.19	-1.17	-1.09	0.43	0.1	-0.57	-0.31	log_dvol_63 z=-0.9; amihud_21 z=+0.2; mom_12_1 z=-1.2	nan
FIS	Technology	mid	0.02	0.02	0.02	-1.12	-0.74	-1.84	-0.28	0.64	-1.17	0.14	rs_sec_63 z=-1.5; clv_21 z=-2.8; log_dvol_63 z=-0.3	nan
LVS	Consumer Cyclical	mid	0.02	0.01	0.01	-0.98	-0.81	-2.47	-1.22	0.82	-1.15	0.8	base_depth_126 z=-1.9; sharpe_21 z=-3.6; log_dvol_63 z=-0.5	nan
FOXA	Communication Services	low	0.02	0.05	0.17	-0.62	-0.02	-0.79	-0.63	0.22	-1.34	-0.58	days_since_2x_low z=+0.6; days_since_20pct_dd z=-0.9; nr7_count_20 z=+0.2	nan
SYK	Healthcare	low	0.02	0.04	0.04	-1.67	-0.99	-2.54	1.18	1.4	-0.28	1.3	days_since_20pct_dd z=-2.2; nr7_count_20 z=+0.8; dvol_trend_21_126 z=+0.1	nan
AON	Financial	low	0.02	0.04	0.05	-0.96	0.5	-2.98	-0.4	1.84	0.16	1.12	days_since_20pct_dd z=-1.6; dvol_trend_21_126 z=+1.2; nr7_count_20 z=+0.9	nan
FOX	Communication Services	low	0.01	0.03	0.26	-0.59	-0.12	-0.76	-0.58	-0.66	-1.44	-0.68	days_since_2x_low z=+0.6; nr7_count_20 z=+1.9; days_since_20pct_dd z=-1.0	nan
ACGL	Financial	low	0.01	0.02	0.51	-0.16	0.08	-0.35	-0.42	4.27	0.6	0.73	days_since_2x_low z=+0.5; dvol_trend_21_126 z=+3.8; nr7_count_20 z=+1.1	nan
DTE	Utilities	low	0.01	0.02	0.31	-0.96	-0.47	-2.99	-0.31	1.83	0.13	2.08	days_since_20pct_dd z=-2.0; dvol_trend_21_126 z=+1.1; streak_now z=+1.0	nan
MGM	Consumer Cyclical	mid	0.01	0.01	0.05	1.11	0.23	-2.6	0.48	0.84	-0.53	2.89	base_depth_126 z=-3.2; rs_sec_63 z=-2.2; log_dvol_63 z=-3.5	nan
GDDY	Technology	low	0.01	0.01	0.02	-0.83	-0.78	-0.34	0.5	0.64	0.07	0.28	days_since_2x_low z=+0.9; dvol_trend_21_126 z=+0.7; streak_now z=+1.8	nan
EQT	Energy	low	0	0.01	0.16	-0.72	-1.29	-1.49	-0.06	-0.24	0.23	-0.81	days_since_2x_low z=+0.6; days_since_20pct_dd z=-1.0; streak_now z=+1.3	nan
CCL	Consumer Cyclical	high	0	0.01	0.14	-1.74	-0.21	-0.68	0.25	1.3	-2.05	-0.6	mom_12_1 z=-1.7; log_dvol_63 z=-0.1; days_since_52w_high z=+1.9	nan
CTVA	Basic Materials	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	fails liquidity / price eligibility today or no completed session (e.g. CTVA unadjusted spin-off)
FDXF	Industrials	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	insufficient history (71 sessions; 252 needed)
HONA	Industrials	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	insufficient history (58 sessions; 252 needed)
Q	Technology	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	insufficient history (216 sessions; 252 needed)
SKYD	Communication Services	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	insufficient history (0 sessions; 252 needed)
VYLR	Basic Materials	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	insufficient history (0 sessions; 252 needed)

6d. Identity vs state: the names raised

ticker	Sector	beta_bucket	level_score	state_score	state_score_pooled	top_states
DELL	Technology	high	0.95	0.9	0.9	mom_12_1 z=+2.3; amihud_21 z=-2.9; log_dvol_63 z=+2.2
NVDA	Technology	high	0.86	0.28	0.25	log_dvol_63 z=-0.8; days_since_52w_high z=-1.1; dn_vol_63 z=-0.6
LLY	Healthcare	mid	0.54	0.24	0.27	rs_sec_63 z=-1.2; corr_spy_63 z=-2.0; sharpe_21 z=-0.6
PGR	Financial	low	0.46	0.77	0.75	nr7_count_20 z=-1.1; days_since_20pct_dd z=-0.7; dvol_trend_21_126 z=-0.3
BRK-B	Financial	low	1	0.84	0.8	days_since_20pct_dd z=+0.6; streak_now z=+1.1; nr7_count_20 z=-0.9
AMD	Technology	high	0.96	0.6	0.63	mom_12_1 z=+1.4; days_since_52w_high z=-1.0; up_day_frac_63 z=+0.7
PANW	Technology	high	0.97	0.85	0.86	log_dvol_63 z=+2.5; amihud_21 z=-1.7; mom_12_1 z=+0.8
NTAP	Technology	high	1	0.98	0.96	log_dvol_63 z=+2.6; mom_12_1 z=+1.0; days_since_52w_high z=-1.1
DUK	Utilities	low	0.93	0.7	0.69	days_since_20pct_dd z=+0.6; nr7_count_20 z=+1.8; streak_now z=-0.5
KO	Consumer Defensive	low	0.96	0.2	0.2	nr7_count_20 z=+1.6; dvol_trend_21_126 z=-0.0; streak_now z=+0.3

Rank correlation between level and state scores across the universe: 0.68.

7. Ticker identity vs. behaviour: how much of the signal generalises across tickers

Each reading is split, point-in-time, into **persistent** (the ticker's own expanding-window mean up to t-1: "this ticker is usually like this") and **deviation** (today's reading as a z-score against the ticker's own history: "this ticker is currently unusual"). IC of each component vs 42d SPY-excess Sharpe:

| factor | recent|raw | recent|persistent | recent|deviation | full|raw | full|persistent | full|deviation | |:------
-----|-----:|-----:|-----:|-----:|-----:|-----:|-----
-----:| | comp_bucketed | 0.044 | 0.049 | 0.031 | 0.013 | 0.015 | 0.014 | | corr_spy_63 | 0.077 | 0.055 |
0.058 | 0.028 | 0.033 | 0.007 | | days_since_20pct_dd | -0.008 | -0.043 | 0.021 | 0.009 | 0.006 | 0.008 | |
dist_52w_high | -0.004 | -0.032 | 0.005 | 0.009 | 0.015 | -0 | | dryup_x_tight | -0.003 | -0.015 | 0.001 | 0.006
| 0.018 | 0.001 | | idio_vol_63 | 0.006 | 0.028 | -0.018 | -0.024 | -0.008 | -0.017 | | mom_12_1 | 0.039 | 0.019 |
0.044 | 0.007 | 0.022 | 0.003 | | obv_price_div_63 | 0.01 | 0.023 | 0.007 | 0.006 | 0.01 | 0.004 | | rs_lead_126
| 0.068 | 0.067 | 0.031 | 0.004 | 0.032 | -0.013 | | sharpe_126 | 0.016 | 0.009 | 0.012 | 0.003 | 0.021 | -0.006
| | up_capture_63 | 0.073 | 0.064 | 0.028 | 0.014 | 0.021 | -0.007 | | vol_dry_20_250 | -0.01 | -0.021 | -0.009
| 0.001 | -0.018 | -0 | | z_mom_12_1 | 0.026 | 0.03 | 0.023 | -0.005 | 0.006 | -0.001 | | z_obv_price_div_63 |
0.003 | 0 | 0.004 | 0.003 | 0.003 | 0.002 | | z_rs_lead_126 | 0.013 | 0.043 | 0.007 | -0.017 | 0.021 | -0.02 | |
z_rv20_pct_252 | 0.004 | 0.044 | 0.001 | -0.007 | 0.011 | -0.008 | | z_vol_dry_20_250 | -0.005 | 0.007 |
-0.007 | -0 | -0.005 | 0.001 |

Pure ticker-identity predictors (long trailing Sharpe; the ticker's own trailing-year mean realised excess Sharpe):

| predictor | ic|full | ic|recent | t_nw|full | t_nw|recent | |:------|-----:|-----:|-----
-:|-----:| | past_xs_1y_mean | 0.015 | 0.044 | 0.927 | 1.699 | | sharpe_252 | 0.013 | 0.042 | 0.758 |
1.615 | | sharpe_504 | 0.013 | 0.026 | 0.651 | 0.865 | | sharpe_756 | 0.024 | 0.034 | 1.266 | 1.303 |

Reading:

- The out-of-sample composite's recent IC (0.047) is matched by its persistent component alone (0.053); the deviation component carries 0.035. Today's composite rank correlates 0.481 with the ticker's own historical-average composite rank. **Roughly half to two thirds of what the ranking expresses is which tickers are chronically strong and stable, not a current state change.**
- A ticker's own trailing-year mean realised excess Sharpe predicts its next 42 days with IC 0.044 (t 1.7) in the recent window, about the same as the whole system. In 2022-10 to 2026 leadership persisted at the ticker level; much of the system's recent edge is that persistence.
- The persistent component is the part most exposed to survivorship bias (today's constituents are disproportionately the names that were chronically strong). Treat the identity half of the score as regime- and universe-dependent.
- The deviation component is the cleaner "behaviour predicts across tickers" evidence. It is weaker but real: 12-1 momentum unusually high for the name (IC 0.044, t 2.0, the only reading where the state component beats the identity component), RS-line leadership unusual for the name (0.031), correlation to SPY unusually high (0.058), idiosyncratic vol unusually low (-0.018). Proximity to 52w high, time since 20% drawdown and the dry-up interaction have no deviation signal pooled: they work only as identity, and only inside beta buckets.
- Volume factors: `obv_price_div_63` and `vol_dry_20_250` show up as identity, not state, pooled. The names that habitually have volume leading price, or habitually dry up, do better; a dry-up that is unusual for the name does not.

7b. Event study: the price / volume / volatility STATE that preceded top-decile forward windows

Episode = a (date, name) whose forward blended SPY-excess-Sharpe percentile was ≥ 0.90 .

Consecutive episode days of one name form a cluster (9,078 clusters since 2014, 2,828 since 2022-10).

The state is measured **5 sessions before the cluster start**: the start day itself is selection-contaminated (by construction the day before was not a top-decile start, so the start is mechanically a local low;

`z_ret_5` reads $t = -59$ at $t=0$ and ~ 0 at $t=5$, see `event_pre_episode_path_recent.csv`). Each state factor is ranked cross-sectionally per date; `mean_rank_excess` is the mean rank at $t=5$ minus 0.5 with a cluster-robust t ; $\text{lift_top_q} / \text{lift_bottom_q} = P(\text{episode} \mid \text{factor in top / bottom quintile}) / P(\text{episode})$, over all days (no start selection).

Recent window, pooled (top 25 by $|t|$):

factor	mean_rank_excess	t_cluster	n_clusters	lift_top_q	lift_bottom_q
<code>z_beta_252</code>	0.02	3.615	2791	1.114	0.937
<code>z_corr_spy_63</code>	0.018	3.202	2805	1.034	0.964
<code>z_pullback_duration_21</code>	0.017	3.184	2809	1.056	0.955
<code>z_dn_up_vol_asym_63</code>	0.016	2.988	2805	1.073	0.921
<code>z_rs_spy_63</code>	-0.016	-2.874	2805	0.845	1.077
<code>z_days_since_ignition</code>	0.015	2.809	2809	1.156	0.971
<code>z_ret_63</code>	-0.015	-2.648	2805	0.875	1.094
<code>z_sharpe_63</code>	-0.014	-2.561	2805	0.886	1.073
<code>z_base_depth_63</code>	-0.012	-2.298	2805	0.943	1.008

factor	mean_rank_excess	t_cluster	n_clusters	lift_top_q	lift_bottom_q
z_beta_63	0.012	2.263	2805	0.967	0.97
z_dn_vol_63	0.012	2.254	2805	1.079	0.914
z_dvol_trend_21_126	-0.012	-2.21	2802	0.914	1.093
z_rs_dist_high_126	-0.012	-2.161	2802	0.951	1.032
z_vol_turn_off_dryup	-0.011	-2.021	2802	0.943	1.044
z_updown_vol_ratio_50	-0.011	-1.953	2805	0.958	1.075
z_higher_low_cadence	-0.011	-1.934	2804	0.964	1.083
z_base_depth_126	-0.01	-1.904	2802	0.925	0.997
z_vol_confirms_price_21	-0.01	-1.893	2807	0.963	1.014
z_ignite_x_rs	-0.01	-1.887	2805	0.941	1.059
z_dist_52w_low	0.01	1.811	2787	1.023	0.916
z_gap_share_21	0.01	1.804	2809	1.127	1.063
z_mom_12_1	0.009	1.733	2787	1.056	0.918
z_ret_21	-0.009	-1.698	2809	0.95	1.01
z_corr_roc_21	0.009	1.671	2804	0.917	1.045
z_vol_ratio_5_50	-0.009	-1.647	2805	0.952	0.994

Recent window, mid-beta tercile (top 15; the low and high terciles are dominated by `z_beta_*` / `z_corr_spy_*` entries that are an artefact of bucketing on the beta level and are not pre-conditions):

factor	mean_rank_excess	t_cluster	n_clusters	lift_top_q	lift_bottom_q
z_days_since_2x_low	0.046	4.45	631	1.316	0.874
z_pullback_21	0.035	3.632	819	1.093	0.846
z_base_tightness_63	-0.035	-3.387	818	0.841	1.06
z_base_depth_63	0.032	3.3	818	1.077	0.83
z_dist_high_roc_10	0.029	3.094	812	1.039	0.922
z_dn_vol_63	-0.029	-2.901	818	0.881	1.002
z_vol_turn_off_dryup	-0.029	-2.827	817	1	0.994
z_ignition_mult_10	0.026	2.724	818	0.967	0.858
z_sharpe_21	0.025	2.593	819	1.044	0.827
z_gap_share_21	-0.025	-2.516	819	1.017	1.111
z_rv20	-0.024	-2.367	819	0.886	1.062
z_overnight_intraday_vol_20	-0.023	-2.307	819	1.012	1.055
z_pos_in_20d_range	0.022	2.251	819	1.065	0.916
z_up_gap_freq_21	-0.022	-2.22	819	1.026	1.014
z_ret_21	0.02	2.092	819	0.97	0.821

Reading:

- Effects are small ($|\text{rank excess}| \leq 0.02$ pooled, lifts 0.85-1.16) and statistically clear (2,800 independent clusters). There is no dramatic P/V/V signature 1-4 weeks before a top-decile 1-3 month window in S&P 500 names; there is a consistent tilt.
- The tilt is a **quiet pullback in the name**: 63d relative strength and 63d return unusually weak for the name (bottom-quintile lift 1.08-1.09), dollar volume unusually low (1.09), longer than usual since the 21d high (1.06), no recent ignition day (top-quintile lift of `days_since_ignition` 1.16), downside-vol asymmetry

unusually high (1.07), beta and correlation to SPY unusually high. This is 1-3 month mean reversion with a capitulation flavour, and it is the opposite of what the level layer rewards.

- Mid beta adds structure that matches the mandate's hypotheses: 63d base unusually tight (bottom-quintile lift 1.06 vs top 0.84), downside vol unusually low (0.88 top), a recent ignition day (bottom-quintile 0.86), time since 2x-off-low crossing unusually long (1.31).
- Volume dry-up vs 250d, Bollinger / range compression, NR7 count, up/down volume ratio and OBV divergence as STATES: no pre-episode tilt pooled ($|t| < 1.5$). Accumulation-day count tilts slightly negative.
- These are the ingredients the state composite is allowed to pick from; what it actually selects each year is in `composite_state_weights_by_fold.csv`.

8. What to trust, what not to

Direct read of the evidence, in order of reliability.

1. What has out-of-sample support for 21-63d forward Sharpe in this universe

- Relative-strength leadership: `rs_lead_126` (RS line closer to its 6-month high than price is to its own). Walk-forward baseline, recent window: 42d IC 0.068 (NW t 2.4), 63d IC 0.082 (t 2.7); still 0.034-0.041 (t 2.1-2.2) after removing beta. The single most useful reading in the library.
- 12-1 momentum: recent beta-neutral 42d IC 0.048 (t 2.4), sector-excess IC 0.043 (t 2.1). Dead over 2014-2021 (full-window IC 0.005), alive since 2022-10. Consistent with the regime shift the mandate describes.
- Beta-bucketed walk-forward composite: recent 42d IC 0.047 (t 1.5), 63d 0.065 (t 1.9), top-minus-bottom decile 42d Sharpe spread +0.29 annualised with near-monotone deciles. The pooled composite has nothing (recent IC negative). Conditioning on beta is what makes the composite work. Its beta-neutral IC is small (0.006-0.015): inside each bucket it still leans on correlation / up-capture type factors, so part of its recent edge is bull-regime beta.
- Low-beta / defensive names respond to volume structure, not price structure: OBV leading price (`obv_price_div_63`, t 3.6 in defensives), volume dry-up vs 250d (`vol_dry_20_250`, t -2.1 in the low-vol tercile), falling dollar volume (`dvol_trend_21_126`, t -2.9 in low beta). Treat as a cluster, not three independent facts.

2. What did not show signal at this horizon in S&P 500 names (tested, not assumed)

- Compression setups (daily-range percentile, Bollinger width, NR7 clusters, base tightness), pooled: IC within +/-0.015 in every window and horizon. They only appear inside the low-vol / defensive bucket. The "compression then expansion" sequence factor is flat everywhere.
- Ignition-day volume multiple, volume z-score, big-volume up-day counts, the ignition x RS interaction: flat. Nothing in a 1-10 day volume burst predicts the next 1-3 months of risk-adjusted return here.
- Accumulation/distribution day counts, up/down volume ratio: flat pooled, mildly negative in places.
- Higher-low cadence, close location value, gap frequency, streaks, 20d range position: flat.

- The 100%-off-low "ceiling": wrong sign pooled and in high beta (names already 2x off the low kept producing better Sharpe, t +2.3 to +2.7); right sign only in Consumer Defensive (t -3.2). Do not use it as a universal exclusion.
- The LightGBM case-study model, in all five configurations: recent-window 42d IC between -0.001 and +0.023, none significant, every variant below the two single-factor baselines. A great 2019 (IC 0.11-0.14) and nothing stable since. Non-linear interaction mining over 86 factors does not beat two robust factors at this horizon with ~10 years of 500-name data. The analog columns in the ranking are therefore descriptive evidence per name, not a forecast.

3. Statistical honesty

- Identity vs behaviour (section 7): about half of the composite's recent IC is reproduced by the ticker's own historical average of the score. That half is regime- and survivorship-exposed. The generalising half is led by momentum / RS leadership unusual for the name.
- No single factor passes a 5% false-discovery-rate test across the 86-factor zoo in either window. The recent window has roughly 24 independent 42-day periods; |t| of 2-2.7 is the ceiling a true IC of 0.05-0.08 can reach there. The claims above rest on (a) coherent clusters pointing the same way, (b) survival after beta-neutralisation, (c) walk-forward reproduction, not on any one t-stat.
- Multiple-testing in the conditional tables: 860 bucket tests (expected max |t| under the null about 3.2) and 946 per-sector tests. Cells near 3 are suggestive only.
- Survivorship: today's constituents. Ranks are robust to it; absolute forward-Sharpe levels (the deciles all sit near -1.1 in the recent window because SPY's own realised Sharpe was high) are not the point.
- Variant selection (5 ML configurations, 2 composite variants) is a mild in-sample choice and is disclosed; the picked composite is the beta-bucketed one.

4. How to use the ranking

- `final_score` is the within-beta-bucket composite percentile, so the top of the list mixes low-, mid- and high-beta leaders by design. Read it with `beta_bucket` : high-beta names are there for trend persistence (RS leading, 12-1 momentum, proximity to highs, recent 2x-off-low crossings); low-beta names are there for quiet volume-led accumulation.
- Expected magnitude: a top-decile vs bottom-decile difference of roughly 0.3 annualised Sharpe over the next 42 days, with the top decile beating the bottom in about 57% of windows. That is a real but modest edge; it is a screen that concentrates probability, not a signal that resolves individual names.
- Refit cadence: the factors turn over slowly (21d rank autocorrelation 0.7-0.9 for the ones that matter), so re-scoring weekly is sufficient; the walk-forward re-estimates weights annually and that was enough.

9. Files

file	content
factor_ic_summary.csv / factor_ic_detail.csv	single-factor IC, NW t, FDR flags, decile spreads, autocorrelation, all horizons / benchmarks / windows

file	content
factor_ic42_by_year.csv	yearly mean IC (42d, SPY-excess) per factor
conditional_ic_{21,42,63}.csv	IC by beta tercile, vol tercile, sector group
conditional_ic_42_per_sector.csv	IC per GICS-style sector
sector_neutral_ic.csv	IC of sector-demeaned ranks vs sector- and SPY-excess Sharpe
factor_rank_corr_recent.csv	factor rank-correlation matrix (recent window)
ml_oos_evaluation.csv , ml_variants_oos_evaluation.csv , ml_oos_ic42_by_year.csv	walk-forward model results
composite_oos_evaluation.csv , composite_weights_by_fold.csv	walk-forward composite results and the factors it chose each year
composite_final_weights.csv , ml_final_feature_importance.csv	what the live score is built from
identity_decomposition_level.csv , identity_decomposition_state.csv , identity_pure_predictors.csv	ticker-identity vs own-history-deviation split of the signal
composite_state_oos_evaluation.csv , composite_state_weights_by_fold.csv , composite_state_final_weights.csv	identity-neutral (state) composite: OOS results, per-year factor picks, live weights
event_pre_episode_profile.csv , event_pre_episode_path_recent.csv	event study: state 5 sessions before top-decile forward windows; rank path t-20..t-0
universe_scores.csv , universe_scores.html	every ticker scored (state, level, ML, z readings, top states, notes)
current_rankings.csv	the live ranking with analog evidence

10. Reproduce

```

pip install -e . scikit-learn lightgbm pyarrow scipy statsmodels tabulate markdown
python -m pvv_score.data_io          # download
python -m pvv_score.build            # features + state transforms + targets
python -m pvv_score.run_eval         # single-factor
python -m pvv_score.run_conditional # conditional
python -m pvv_score.model            # walk-forward ML
python -m pvv_score.run_model_variants
python -m pvv_score.composite        # walk-forward composite (level)
python -m pvv_score.score_now        # level-based ranking (comparison)
python -m pvv_score.run_identity level
python -m pvv_score.composite --state # identity-neutral composite
python -m pvv_score.run_identity state
python -m pvv_score.events           # pre-episode event study
python -m pvv_score.score_state      # full-universe state scoring
python -m pvv_score.ranking_page     # sortable all-ticker page
python -m pvv_score.report

```